MSc Research Final Presentation, 03rd July 2025

Towards disentangling the interrelationships between hydro-meteorological hazards, conflict and displacement: a case study (for droughts and floods) in Somalia

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Enschede, The Netherlands,
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Thesis submitted to the Faculty of Geo-Information Science and Earth Observation of the University of Twente in partial fulfilment of the requirements for the degree of Master of Science in Geo-information Science and Earth Observation.
Specialization: Geoinformatics

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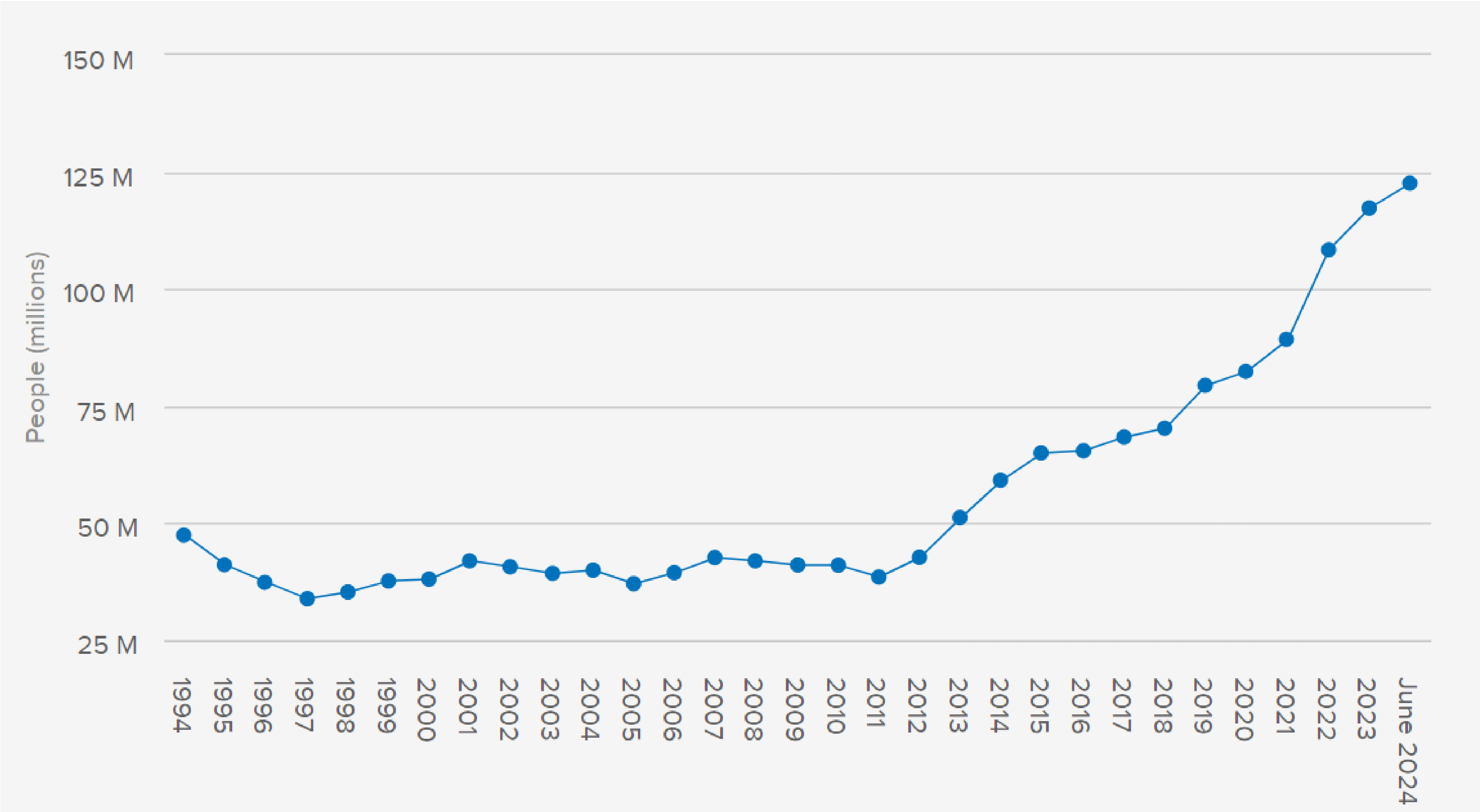
INTRODUCTION

GLOBAL TRENDS FORCED DISPLACEMENT IN 2024

122 million people globally were forcibly displaced

Over 3.8 million Somalis displaced in 2023, up from 1.1 million in 2013.



Source: UNHCR Mid-Year Trends 2024, 9 October 2024.

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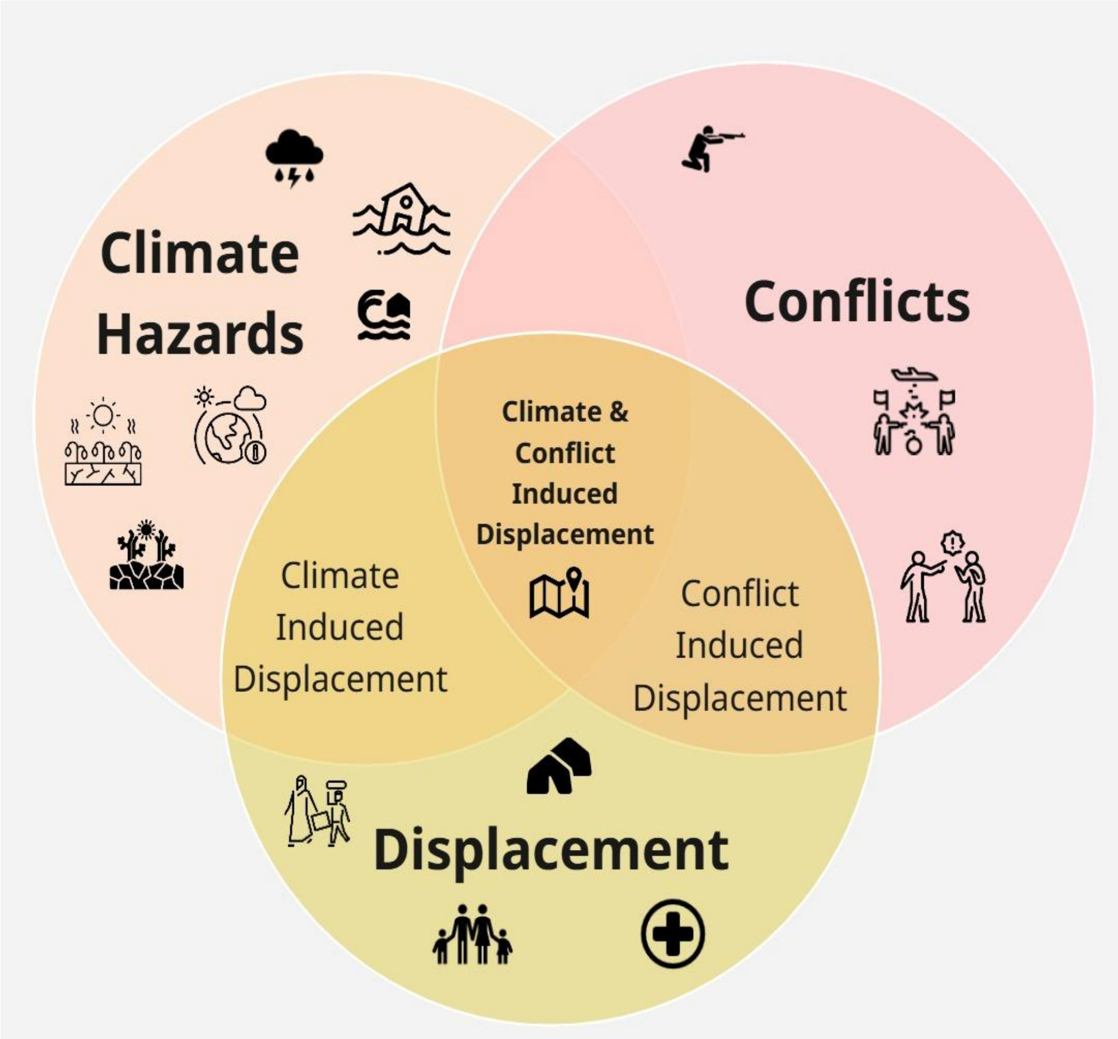
RESEARCH MOTIVATION

- Between 2022–2023:
  - ➢ Somalia faced **prolonged drought, severe floods**, and ongoing conflict led to displacement.

- Existing research often analyse **displacement** triggers **separately**:
  - ➢ **Natural disasters**
  - ➢ **Conflict**

- Research gap: Limited understanding of how conflict and climate hazards **interact** to produce displacement in Somalia



Interrelationships

RESEARCH OBJECTIVE

The main objective of this research is to investigate how **conflict and extreme weather events (drought and floods)** influenced the spatial and temporal dynamics of **displacement** across Somalia in 2022 and 2023.

RO1: To analyse the relationship between internal displacement, conflict, and drought in Somalia during 2022.

RO2: To analyse the relationship between internal displacement, conflict, and flooding during the El Niño-driven events in Somalia in 2023.

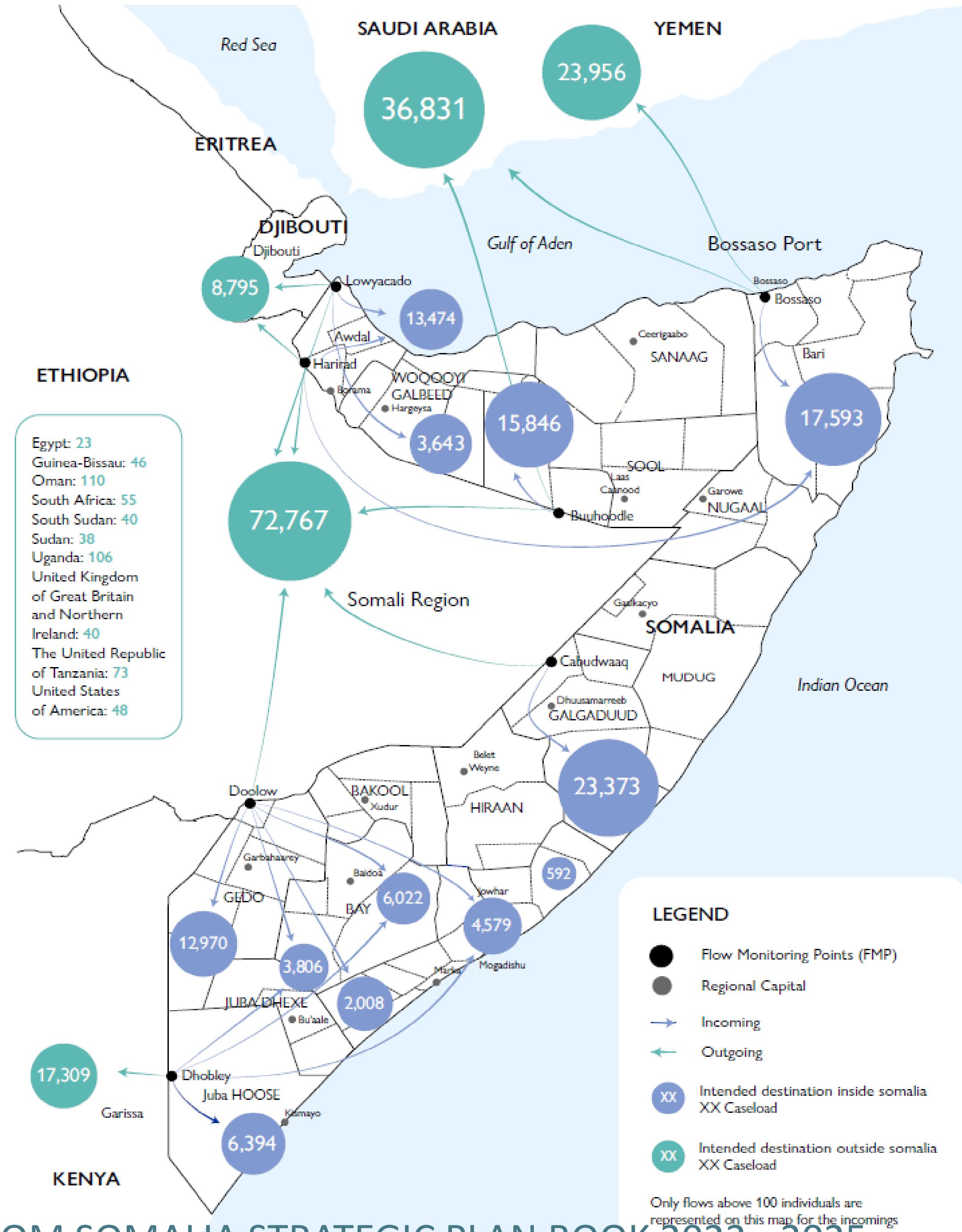
RO3: To evaluate the influence of resolution on the ability to analyse displacement dynamics in relation to conflict and weather events

STUDY AREA

- Study Area: Somalia
- Displacement is not just internal; it links to broader migration systems.
- Somalia is both a source and destination for human mobility.

Datasets Used

- Flood Data: Flood extent
- Rainfall Data: CHIRPS
- Displacement Data: UNHCR-PRMN
- Conflict Data: ACLED (2022–2023)
- Disaster Impact Data: EM-DAT

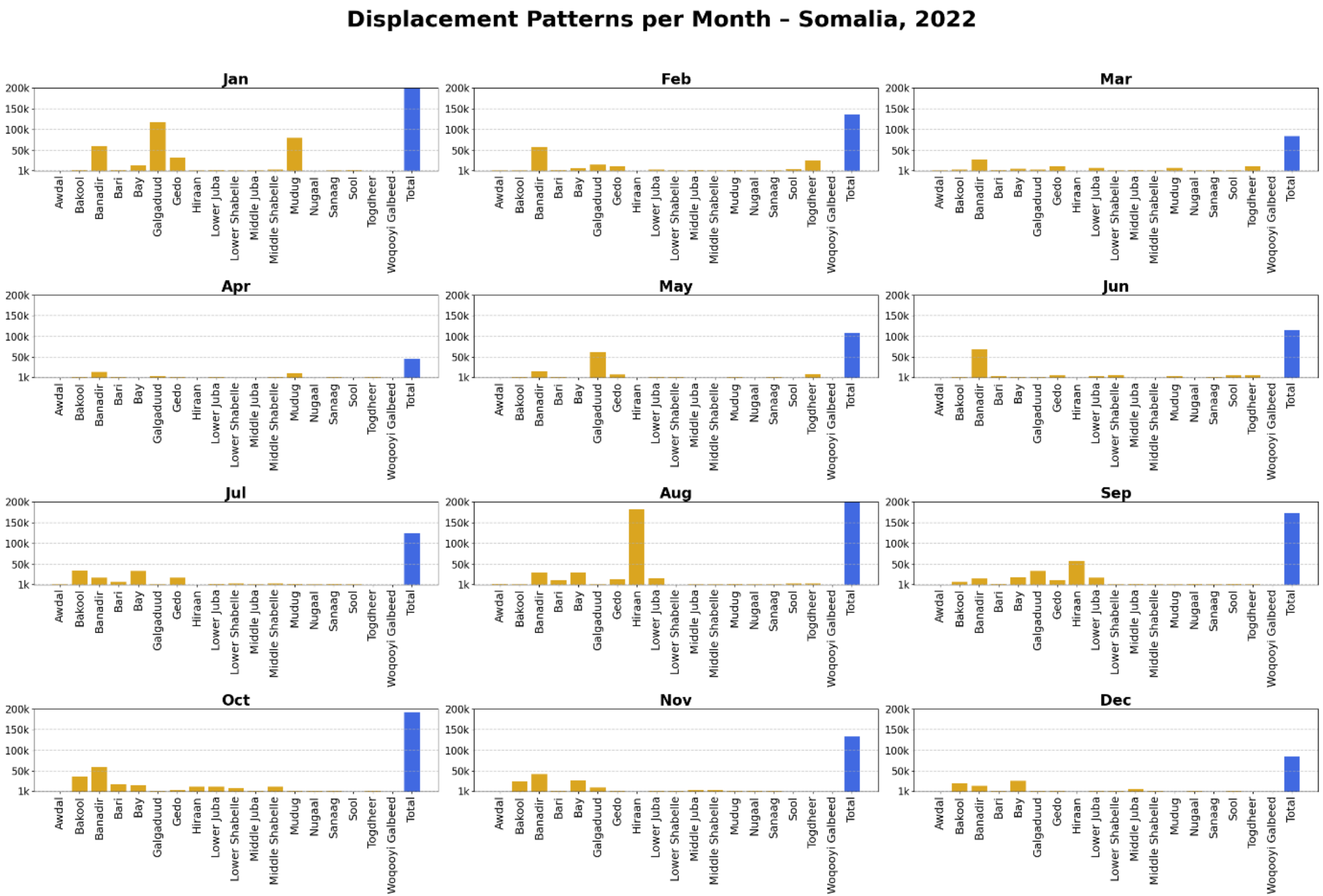


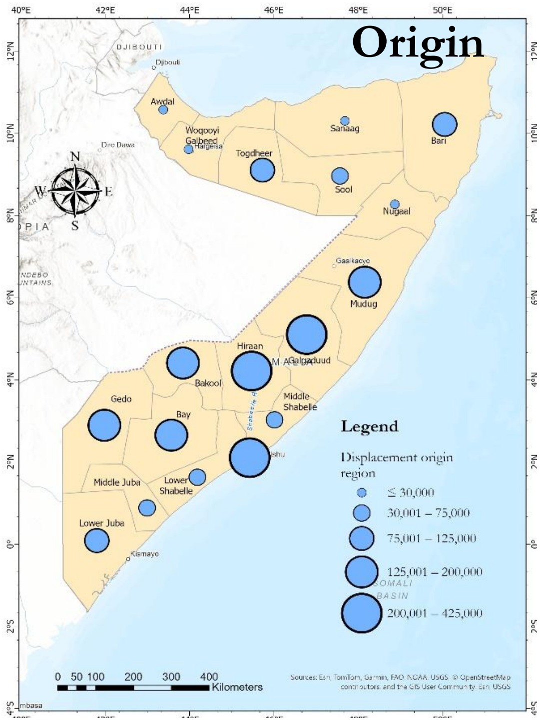
Source of the map: IOM SOMALIA STRATEGIC PLAN BOOK 2022 - 2025

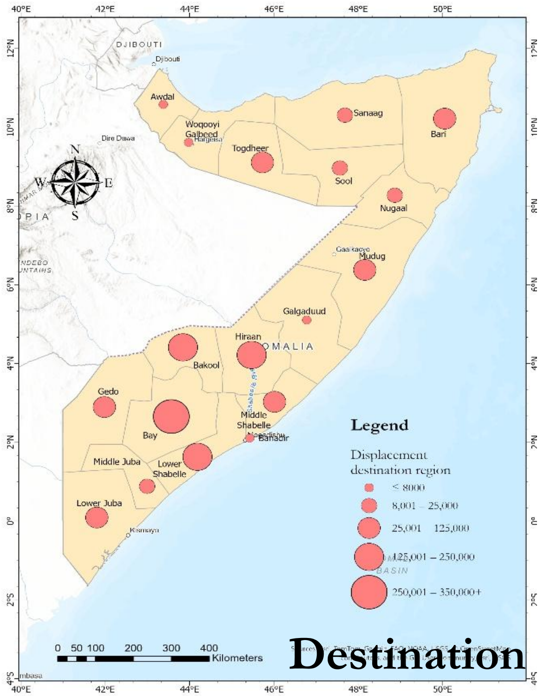
METHODOLOGY

Displacement Analysis	Hazard Analysis	Attribution & Resolution
PRMN Displacement Data • Monthly, Admin 2 • IDP cause + numbers *Data Cleaning & Aggregation*	**Conflict Events (ACLED)** • Filtering by date & fatalities **Drought & Flood Data** • CHIRPS rainfall & SPI maps • SWALIM flood zones → *Buffer*	**Monthly/Admin 2 Harmonization** → *All datasets aligned spatially & temporally* **Displacement Attribution** • Drought: SPI ≤ –1.5, no persistent conflict • Flood: within flood extent buffer • Conflict: >10 events/month • Compound: combination
Output: Monthly Displacement per Region & Month	Output: Conflict Events, Drought/Flood Maps	Output: Displacement Attribution (2022–2023)

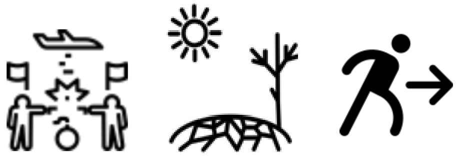
RO1. Displacement Patterns in 2022



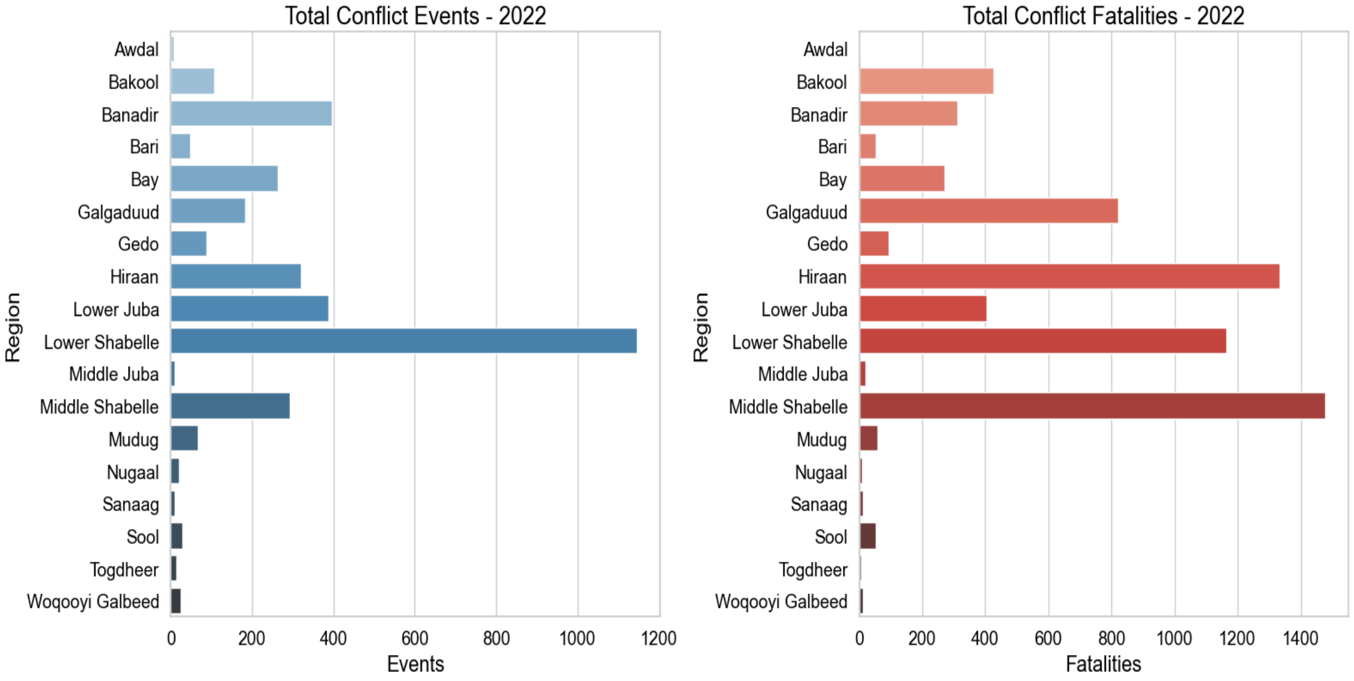


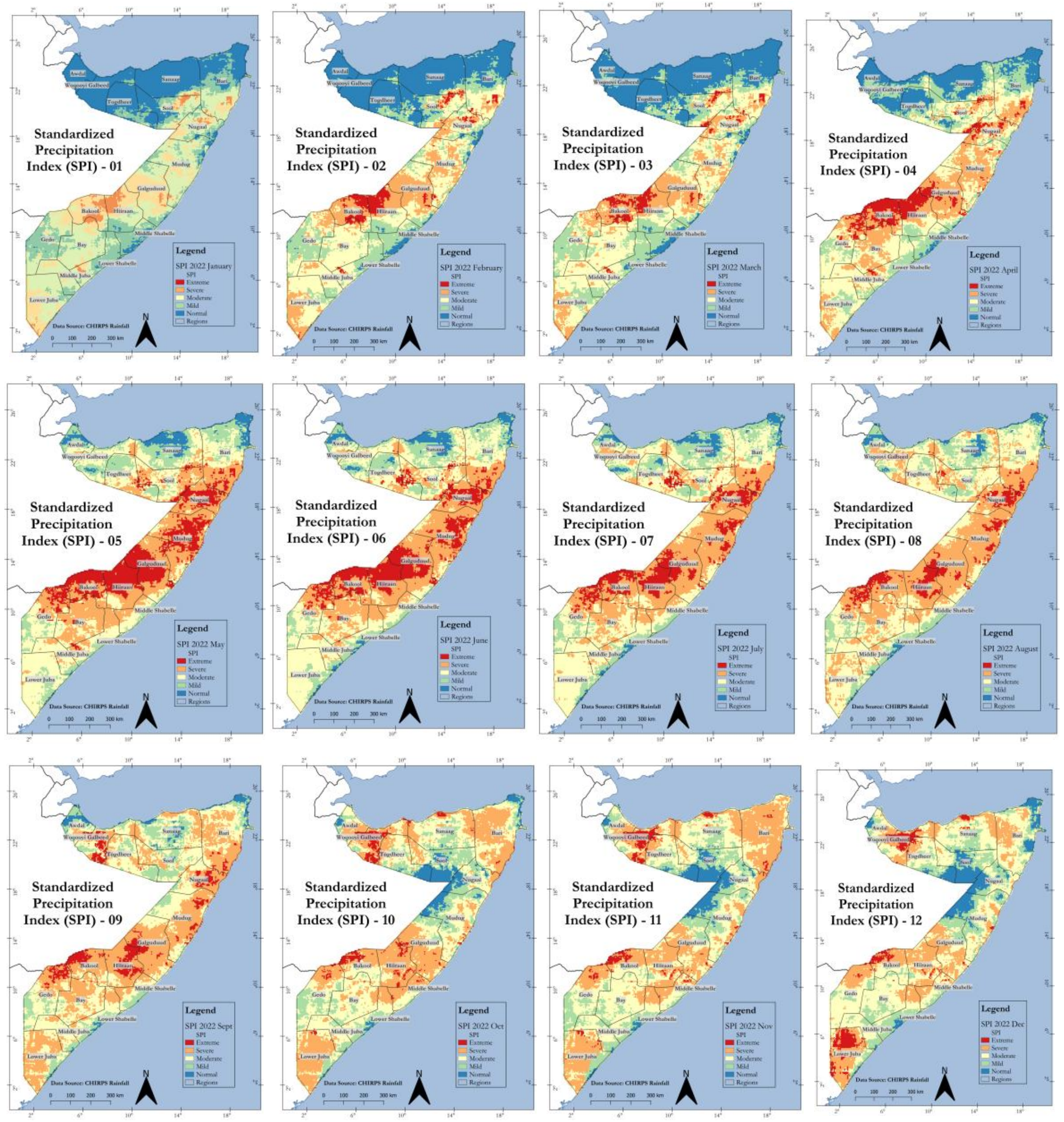


RO1. Conflicts and Drought in 2022



Total Conflict Events and Fatalities by Region (2022)



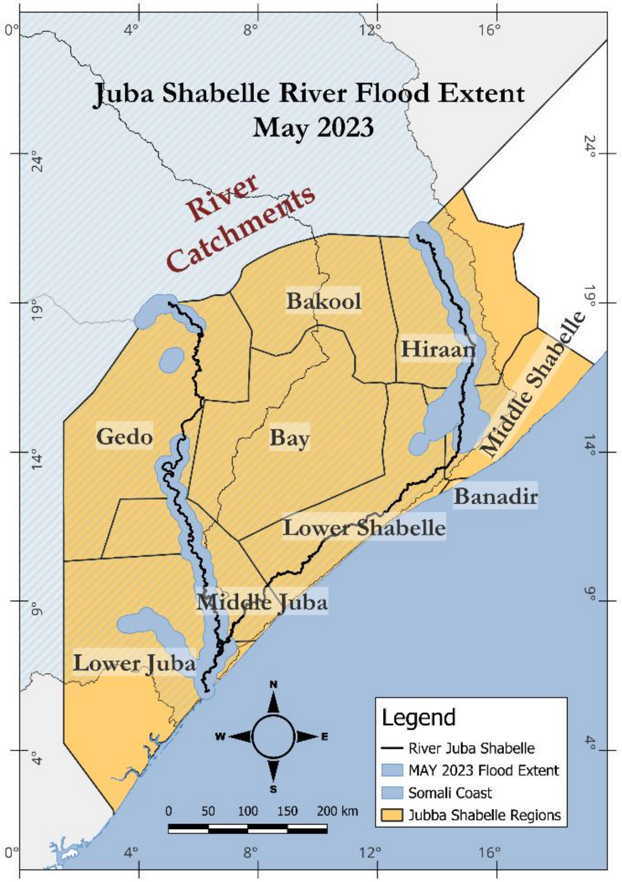


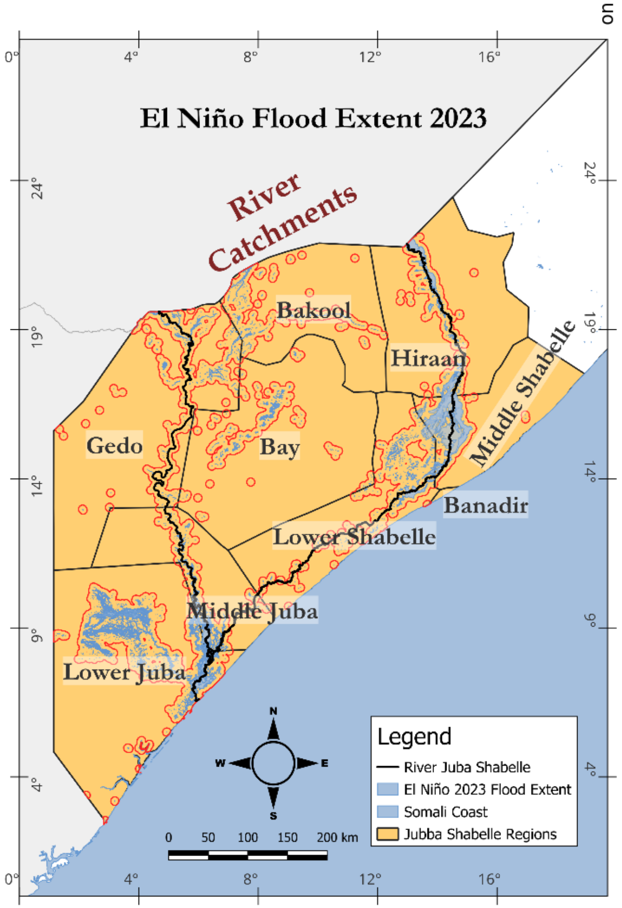
SPI Monthly Drought Maps

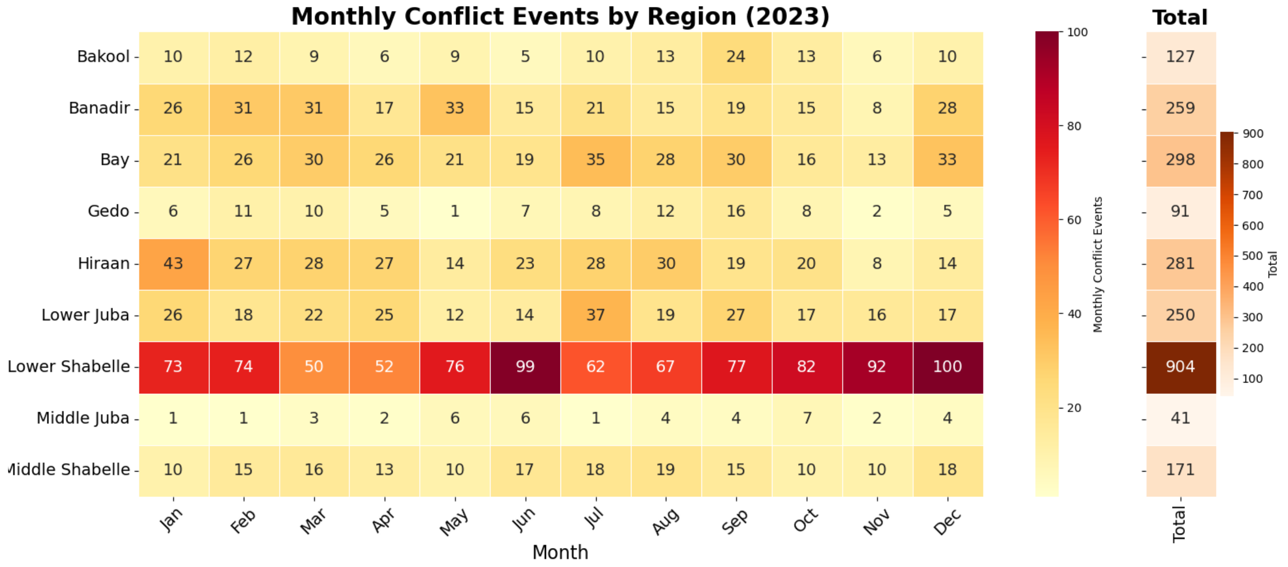
RO1. Displacement attribution for 2022

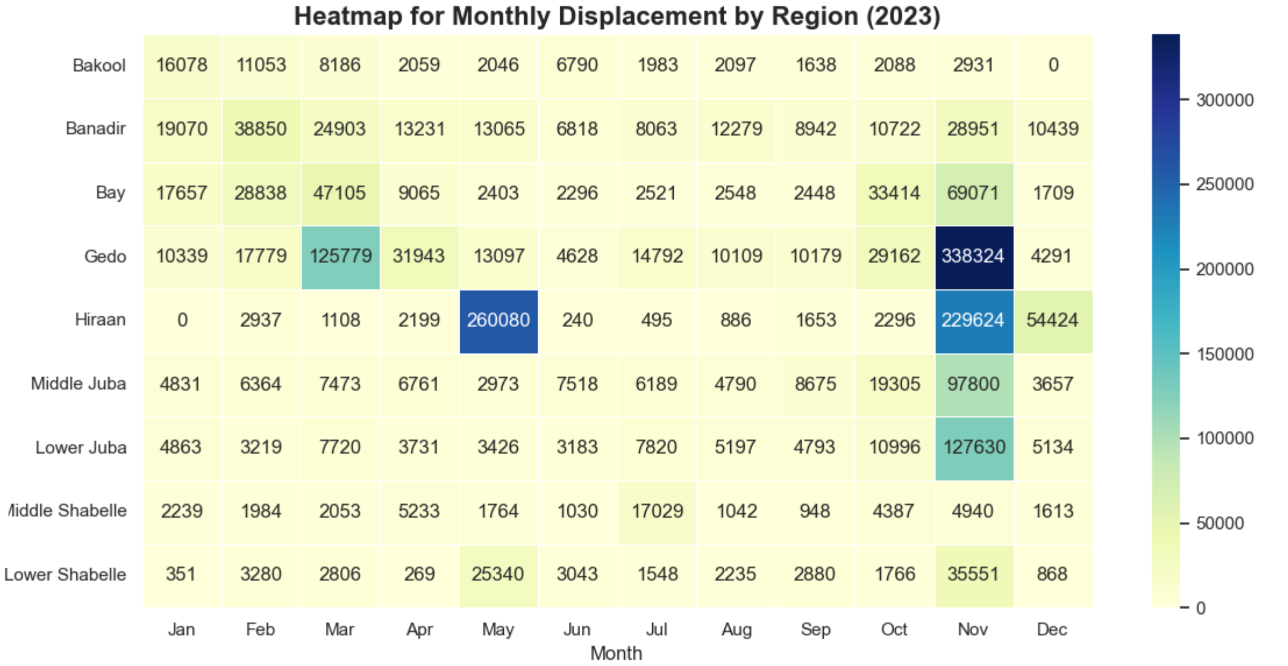
Displacement Type	Condition	Regions
Conflict-aligned	Conflict intensity exceeded threshold, no drought severity (SPI > –1.5)	Middle Shabelle
Drought-driven	SPI ≤ –1.5 (moderate to severe drought), no active conflict	Bari, Bakool (during rainy seasons)
Compound (Conflict + Drought)	Both conflict and drought occurred in same month and region	Lower Juba, Lower Shabelle

RO2. Displacement, conflict and floods









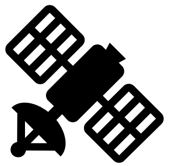
RO2. Displacement attribution for 2023

Displacement Type	Regions	Peak Displacement Figures
Flood-induced	Gedo, Lower Juba	Gedo: 338,324 (Nov) L. Juba: 127,630 (Nov)
Conflict-driven	Bay, Bakool	Bay: 47,105 (Mar)
Compound (Flood + Conflict)	Hiiraan, Lower Shabelle, M. Shabelle, Banadir	Hiiraan: 229,000+ (Nov) L. Shabelle: 92,312 (Nov)

RO3. Data Harmonization

Dataset	Original Spatial Resolution	Original Temporal Resolution	Processed Spatial Resolution	Processed Temporal Resolution
UNHCR PRMN (Displacement)	Region (Admin 2)	Monthly	Admin 2	Monthly
ACLED (Conflict)	Point-based	Daily	Admin 2	Monthly
CHIRPS (Rainfall)	~5 km raster (pixel level)	Daily	Admin 2-level drought interpretation	Monthly (SPI-1)
Flood Extent	Event-based shapefiles	Event-specific	Admin 2 (via spatial overlap)	Months of the event

- All datasets resampled to Admin 2 spatial and monthly temporal resolution
- Consistent classification of displacement events by trigger
- Monthly aggregation smoothed short-duration events like flash floods or brief conflict spikes





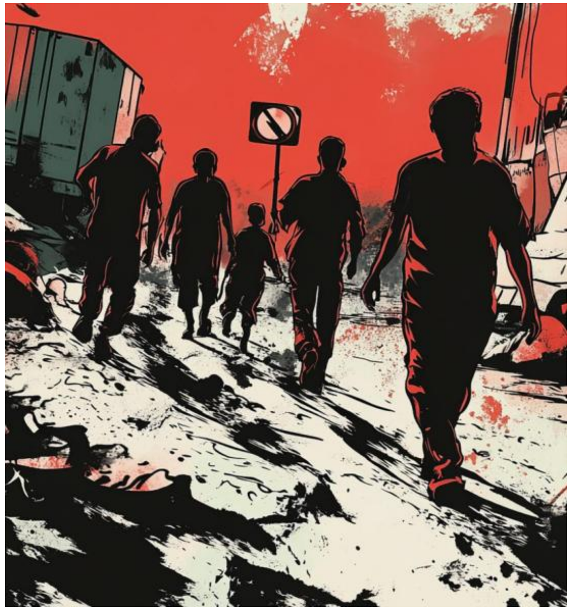
CONCLUSION

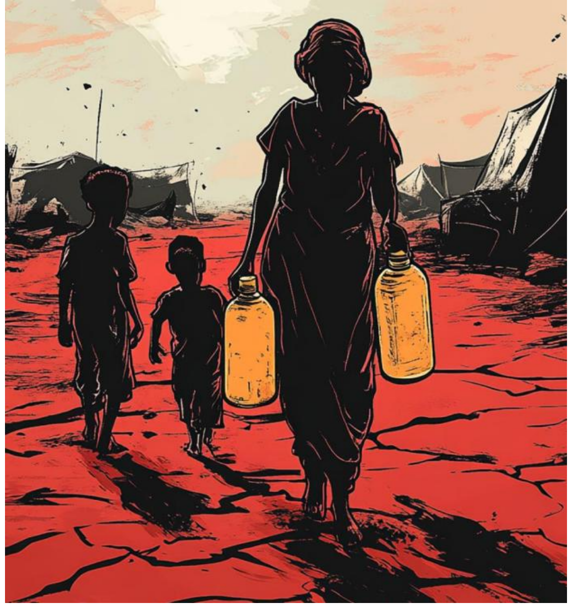
How are hydro-meteorological hazards and conflict events influencing displacement risk in Somalia?

- First study to track 2022 drought and 2023 flood displacement with conflicts
- Drought dominated in 2022; 2023 floods caused major spikes.
- Conflict acts alone and compounds displacement!

Recommendations

- Cross-Border: Displacement across Kenya and Ethiopia.
- Conflict: Analyse severity, violence types, actors.
- Drought: Testing better drought indices.
- Validating Attribution: Use local knowledge (interviews, FGDs, PRMN data)
- Immobility: voluntary vs. involuntary immobility.





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Thank you

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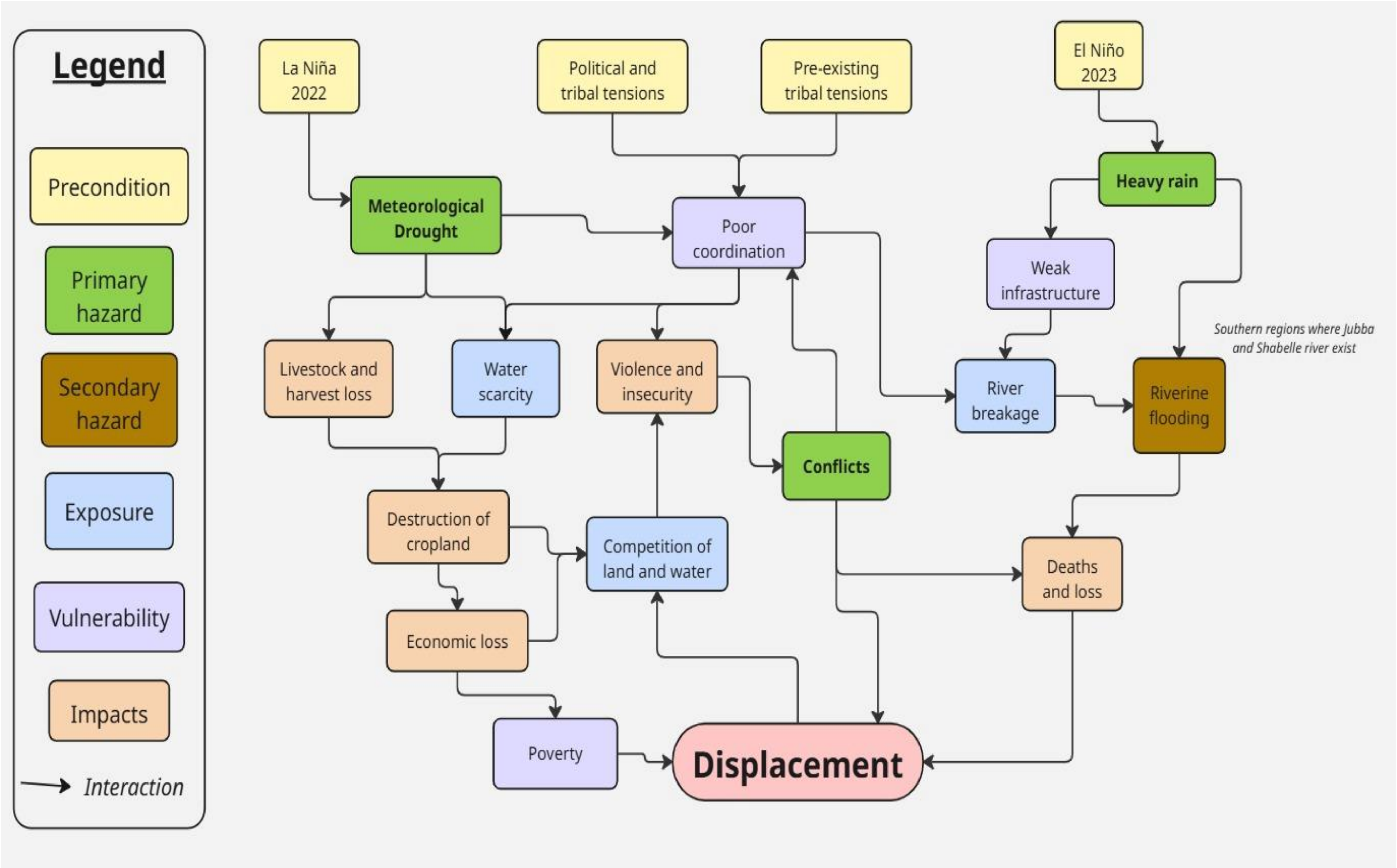
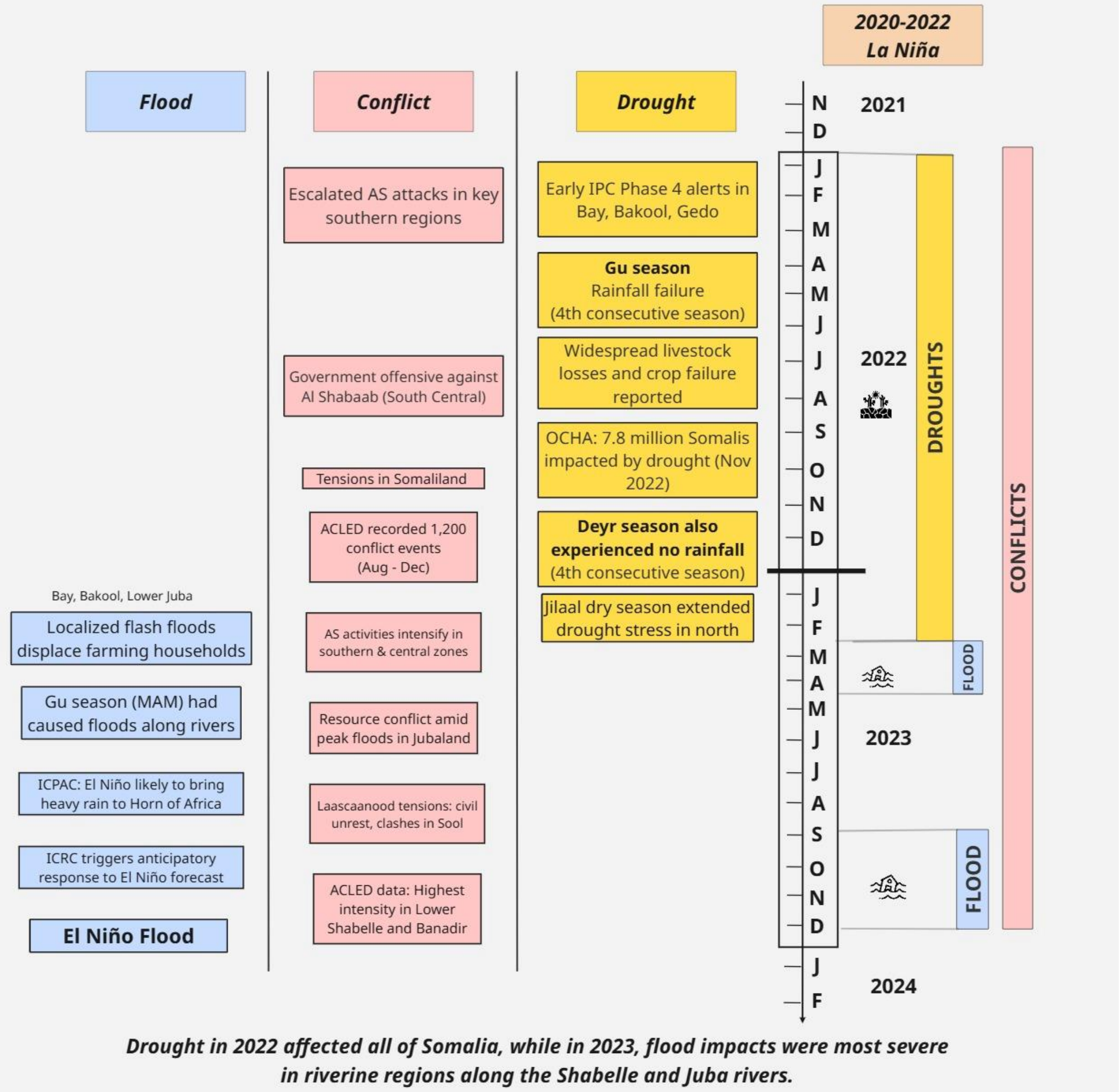
SUPPORTING SLIDES

IMPACT CHAIN



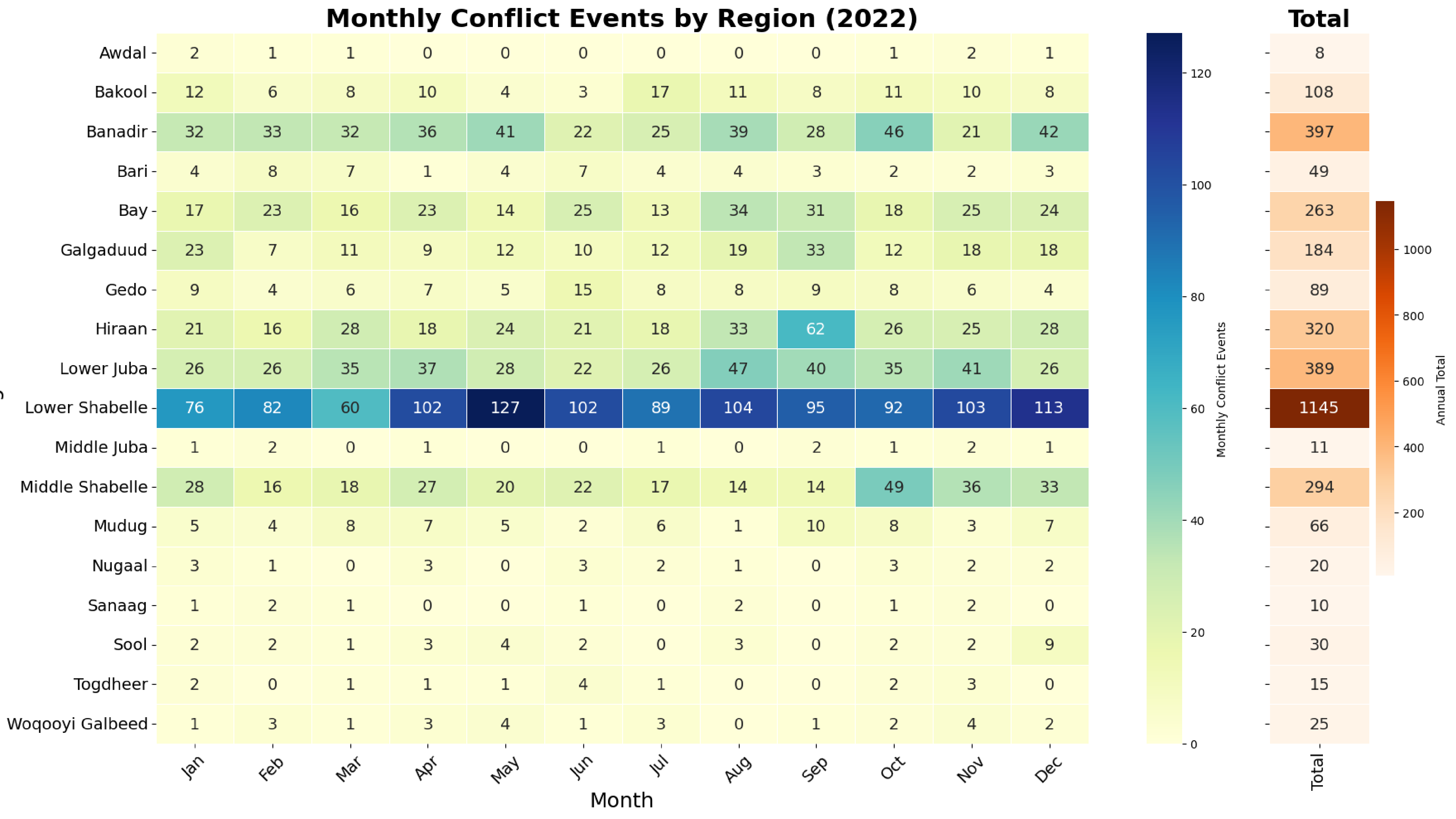
EVENT TIMELINE



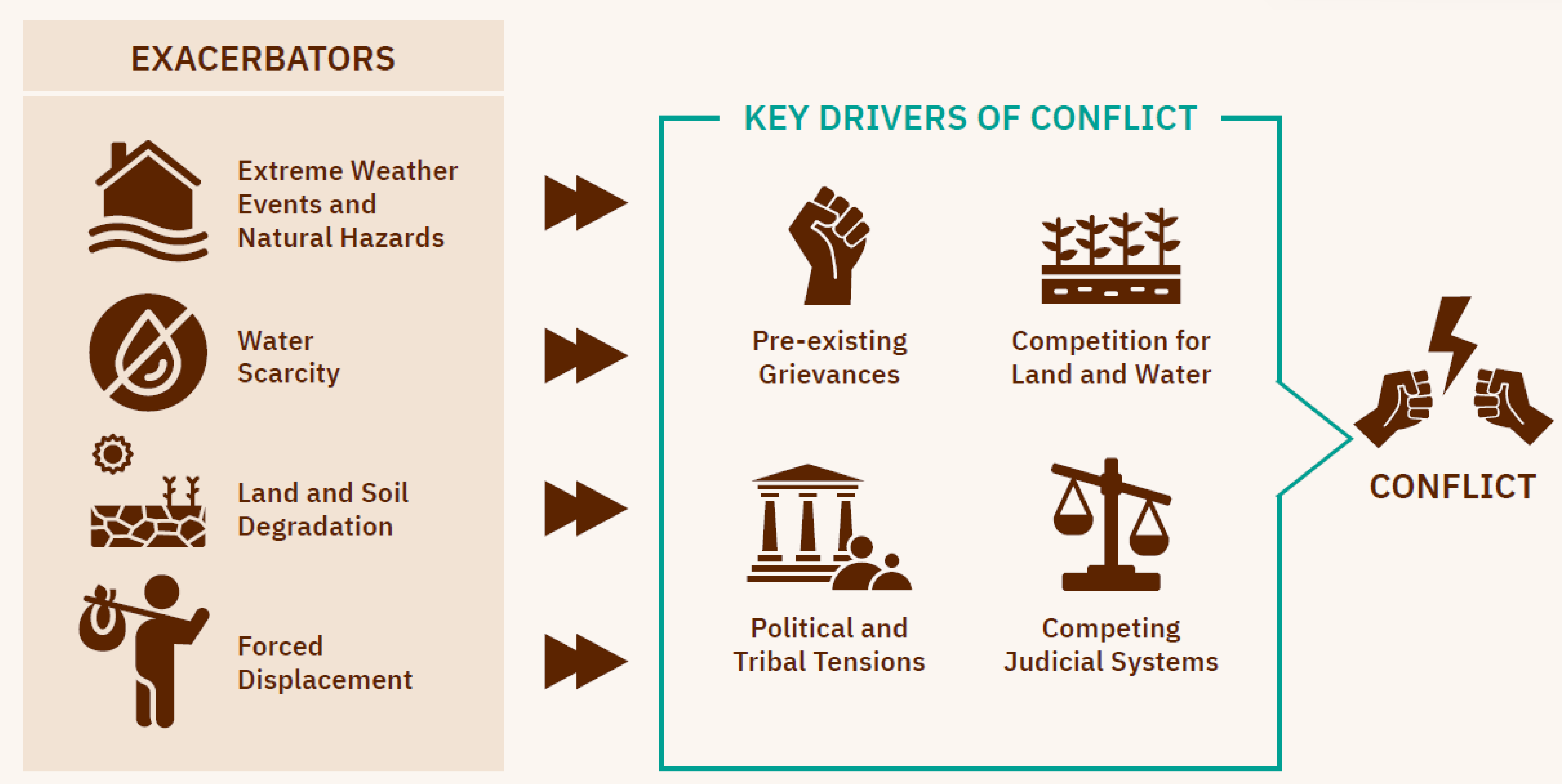
Generic Event Timeline 2022-2023

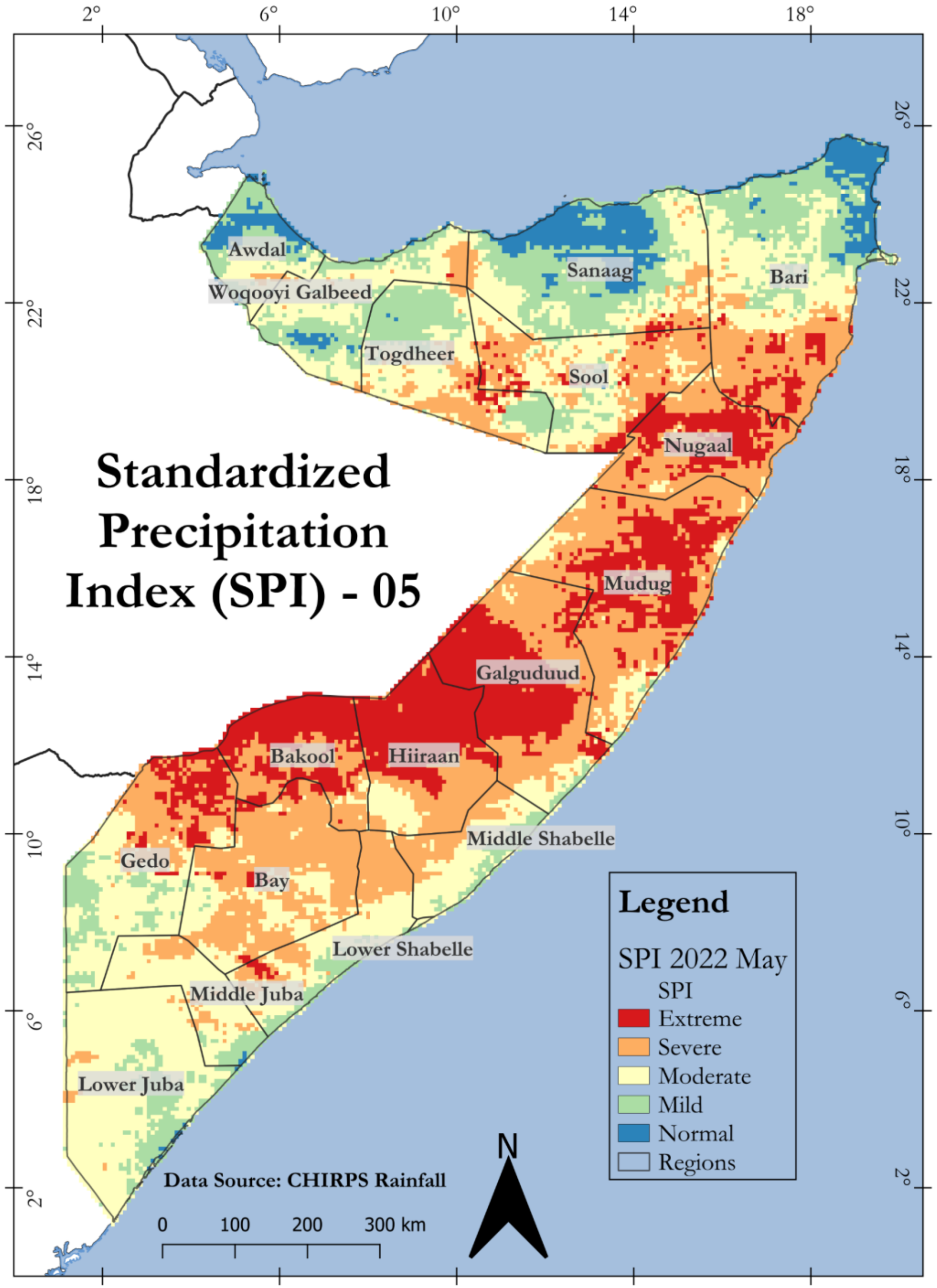
Overview of datasets

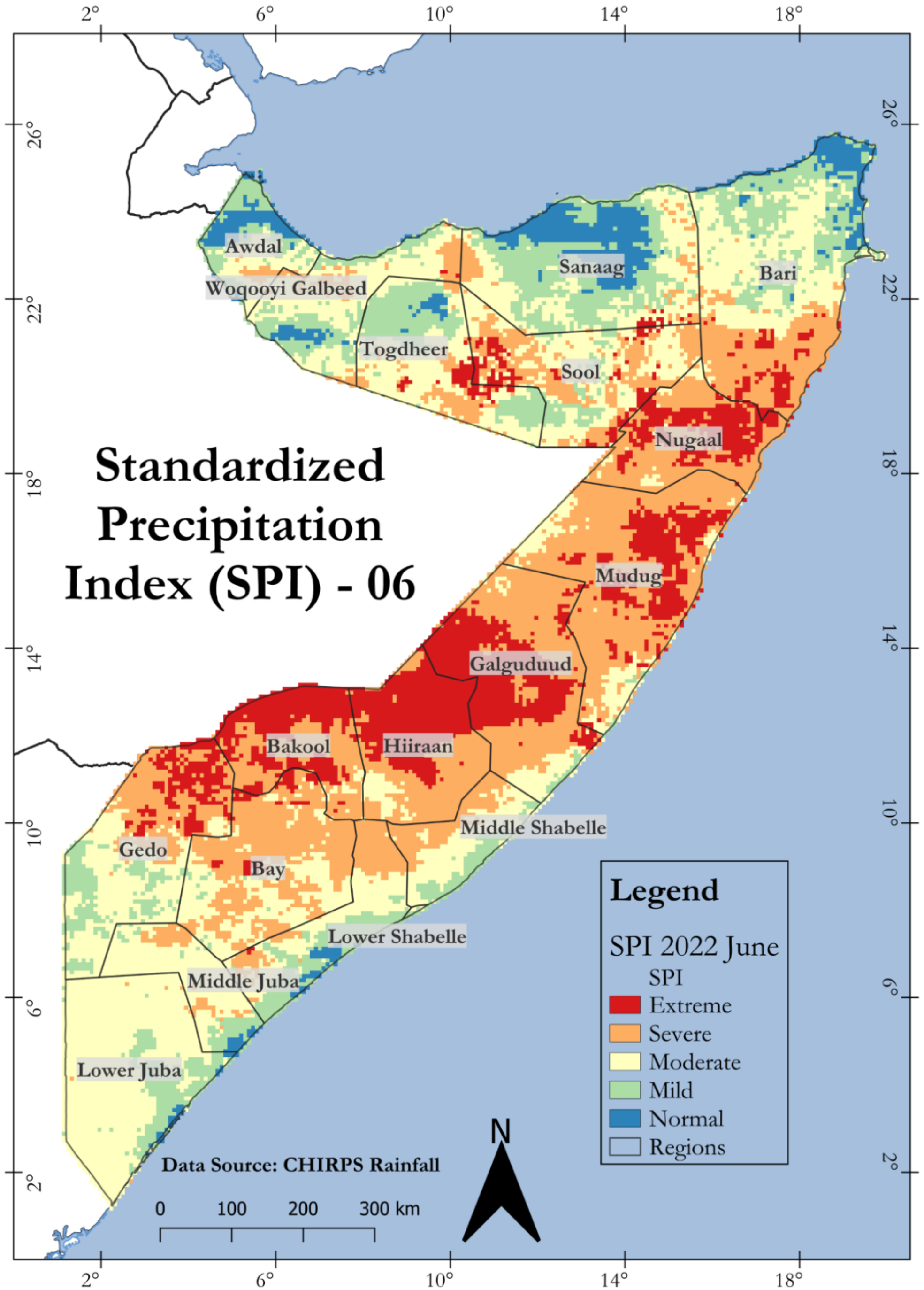
Data Category	*Dataset*	*Spatial Resolution*	*Temporal Resolution*	*Data Type & Format*	*Source of Data*	*Role in study*	*URL*
Administrative boundaries	Country, states, regions, disticts	National to sub-national level	Static	Vector, Shapefile	Global Administrative Areas	RO1 & RO2	Somalia
Historical Flood Event Data	Historical flood	Polygon flood extent	Event based	Raster, GeoTiff	SWALIM	RO2	Juba Shabelle River Floods
Meteorological Data	Rainfall (20 years)	5566 meters pixel size	Daily	GeoTiff	CHIRPS	RO1 & RO3	CHIRPS
Displacement Data	Data on displacement patterns	Aggregated data Admin 2	Weekly to monthly	Tabular, CSV with coordinates.	UNHCR	RO1, RO2 & RO3	PRMN
Conflict	Conflict events recorded in the study period	Point data Admin 2	Daily events 2022 and 2023	Vector, Shapefile	ACLED	RO1, RO2 & RO3	ACLED
Impact	Natural hazard events include the date, disaster type, area and number of affected people	Region Admin 2	Based on event	Excel .xlsx	EM-DAT database	RO1 & RO2	EM-DAT


Monthly Conflict Events by Region (2022)
Total
Awdal 2 1 1 0 0 0 0 0 0 1 2 1 8
Bakool 12 6 8 10 4 3 17 11 8 11 10 8 108
Banadir 32 33 32 36 41 22 25 39 28 46 21 42 397
Bari 4 8 7 1 4 7 4 4 3 2 2 3 49
Bay 17 23 16 23 14 25 13 34 31 18 25 24 263
Galgaduud 23 7 11 9 12 10 12 19 33 12 18 18 184
Gedo 9 4 6 7 5 15 8 8 9 8 6 4 89
Hiraan 21 16 28 18 24 21 18 33 62 26 25 28 320
Lower Juba 26 26 35 37 28 22 26 47 40 35 41 26 389
Lower Shabelle 76 82 60 102 127 102 89 104 95 92 103 113 1145
Middle Juba 1 2 0 1 0 0 1 0 2 1 2 1 11
Middle Shabelle 28 16 18 27 20 22 17 14 14 49 36 33 294
Mudug 5 4 8 7 5 2 6 1 10 8 3 7 66
Nugaal 3 1 0 3 0 3 2 1 0 3 2 2 20
Sanaag 1 2 1 0 0 1 0 2 0 1 2 0 10
Sool 2 2 1 3 4 2 0 3 0 2 2 9 30
Togdheer 2 0 1 1 1 4 1 0 0 2 3 0 15
Woqooyi Galbeed 1 3 1 3 4 1 3 0 1 2 4 2 25
Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec
Month
Total
Monthly Conflict Events
0 20 40 60 80 100 120
Annual Total
200 400 600 800 1000

Pathways between exacerbators and key drivers of conflict in Somalia (EIP, 2024)




Standardized Precipitation Index (SPI) - 05
Awdal
Woqooyi Galbeed
Togdheer
Sanaag
Sool
Bari
Nugaal
Mudug
Galguduud
Bakool
Hiiraan
Middle Shabelle
Gedo
Bay
Lower Shabelle
Middle Juba
Lower Juba
Legend
SPI 2022 May
SPI
Extreme
Severe
Moderate
Mild
Normal
Regions
Data Source: CHIRPS Rainfall
0 100 200 300 km
N
2° 6° 10° 14° 18°
26° 22° 18° 14° 10° 6° 2°

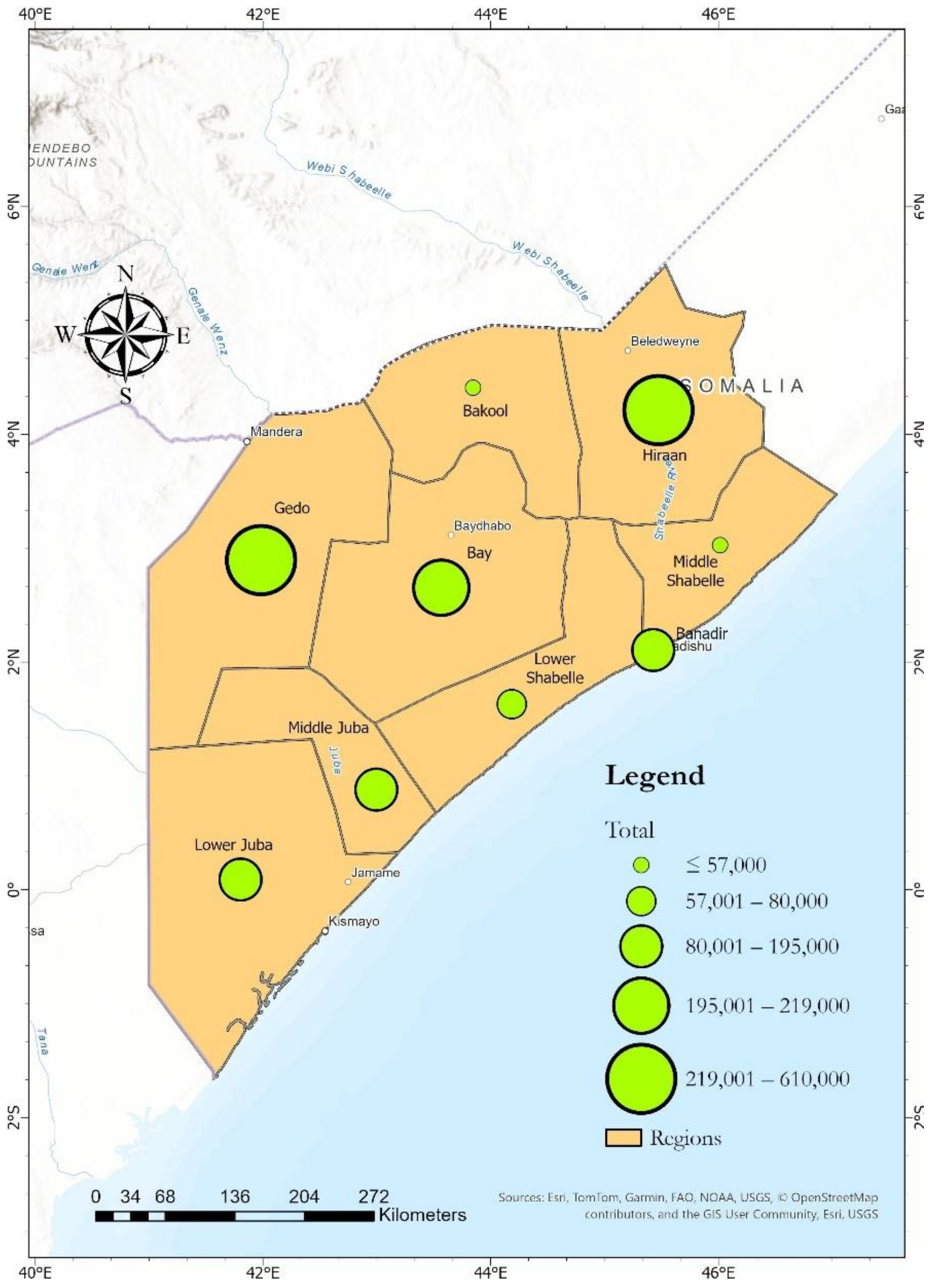

Standardized Precipitation Index (SPI) - 06
Awdal
Woqooyi Galbeed
Togdheer
Sanaag
Sool
Bari
Nugaal
Mudug
Galguduud
Bakool
Hiiraan
Middle Shabelle
Gedo
Bay
Lower Shabelle
Middle Juba
Lower Juba
Legend
SPI 2022 June
SPI
Extreme
Severe
Moderate
Mild
Normal
Regions
Data Source: CHIRPS Rainfall
0 100 200 300 km
N
2° 6° 10° 14° 18°
26° 22° 18° 14° 10° 6° 2°

Displacement Attributed to Conflict and Drought for 2022

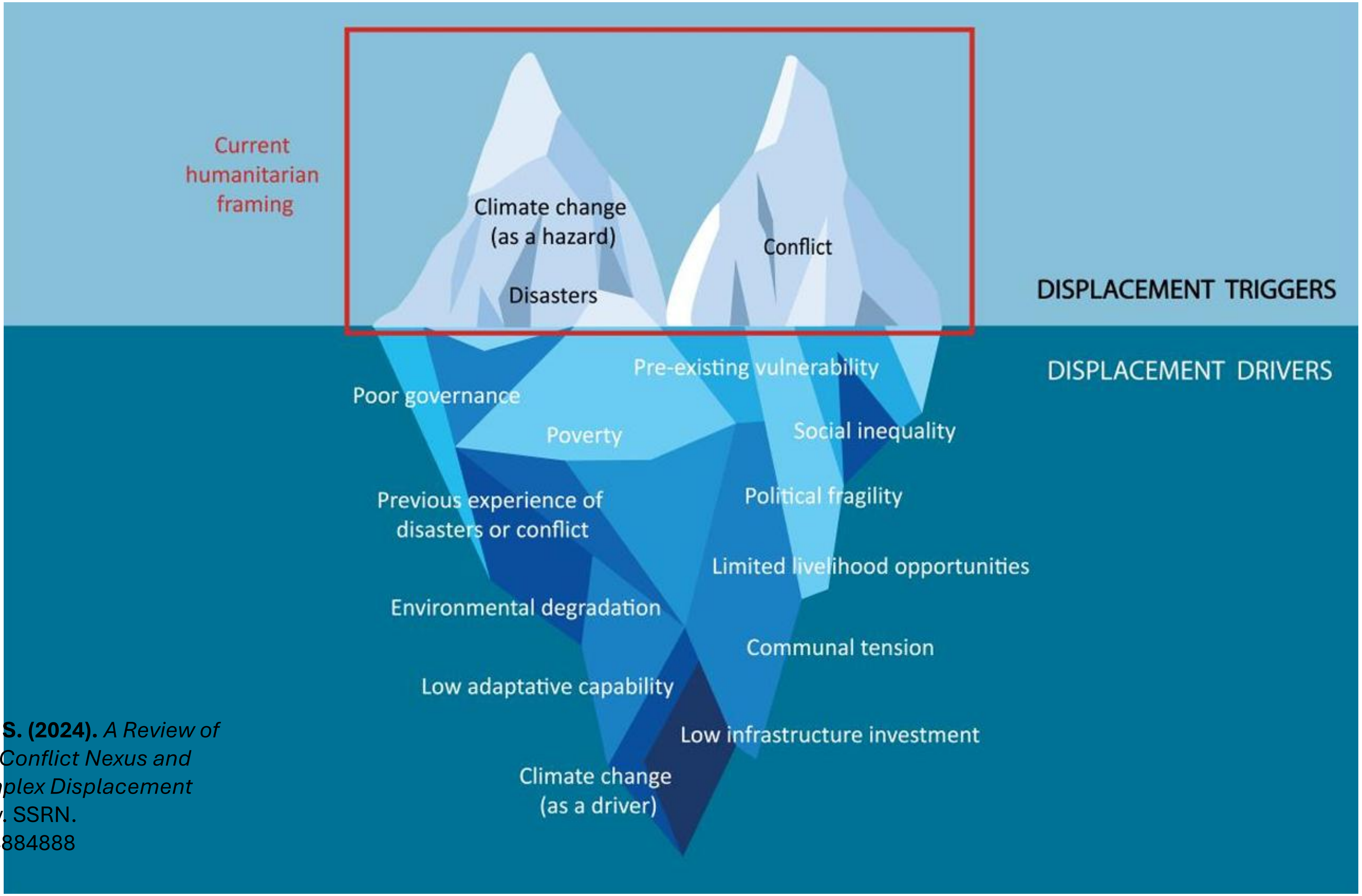
Region	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Awdal	307	698	477	690	1128	897	503	2080	372	660	893	634
Bakool	__2077__	1191	2730	__1128__	2710	1180	__34556__	__1244__	7364	__35943__	__24718__	20432
Banadir	__60141__	__58016__	__27993__	__14744__	__15387__	__68300__	__17033__	__29148__	__14928__	__59277__	__42763__	__14730__
Bari	1522	2261	2194	1898	1903	3944	7804	12096	2371	17275	20537	20131
Bay	__12989__	__2890__	__501__	__745__	_6781_	_2194_	__33440__	__20969__	__2906__	__12065__	__27347__	__26309__
Galgaduud	__117896__	15041	__2682__	3078	1800	__5520__	__17200__	__13789__	__1700__	__1728__	__9847__	1903
Gedo	32150	10976	10159	5925	2989	1606	18226	15794	57288	11850	6000	2509
Hiiraan	_669_	_1098_	_165_	_177_	_700_	_700_	_170_	_266_	_132_	_200_	_200_	_165_
Lower Juba	__1965__	__3629__	__7894__	__1799__	__2766__	__3255__	__2324__	__15888__	__16813__	__12199__	__2900__	2084
Lower Shabelle	__1303__	__1266__	__808__	__634__	__1090__	__1194__	__800__	__1659__	__1629__	__2388__	__1339__	__1350__
Middle Juba	733	2200	2009	1089	582	1028	1072	1085	2288	2285	677	677
Middle Shabelle	__2853__	__1394__	__1414__	__1161__	__1104__	__965__	__2884__	__1419__	__641__	__1129__	__3520__	__1063__
Mudug	79826	2454	7400	11265	10620	12250	12950	15932	_15290_	12059	3063	673
Nugaal	1284	1222	466	536	446	442	452	452	452	452	452	452
Sanaag	1101	902	1340	1200	637	1261	1644	810	619	1130	1139	933
Sool	1919	4088	5007	5124	5164	3970	1991	1657	1692	1629	1870	1570
Togdheer	347	25515	10972	2361	1983	6168	198	2811	1209	2382	667	833
Woqooyi Galbeed	80	111	143	87	77	211	144	178	215	227	233	144

Regions experiencing severe or extreme drought conditions are highlighted in **bold (orange),** while those surpassing the conflict thresholds are expressed with **underlined text (blue).** Regions and months where both conflict and drought thresholds were met simultaneously are displayed in bold and underlined text **(yellow).**

Displacement per Region (2023)

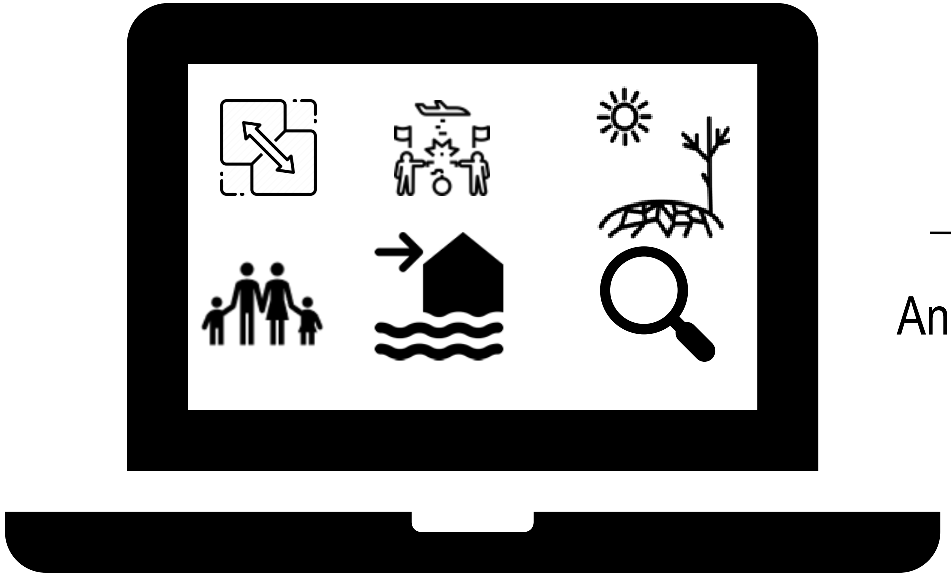


Current humanitarian organisation framing and labelling of complex displacement contexts



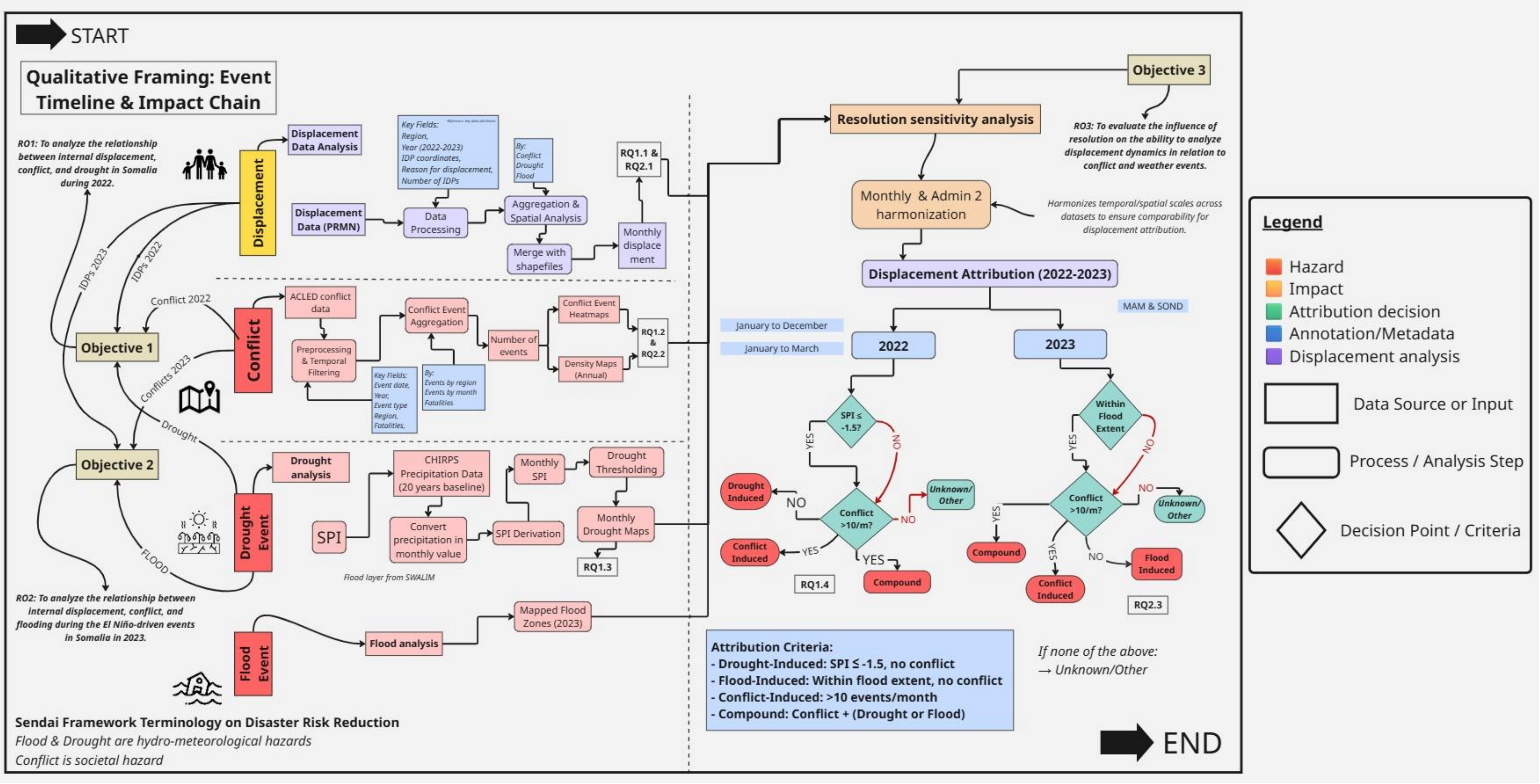
See, S., Opdyke, A., & Banki, S. (2024). *A Review of the Climate Change-Disaster-Conflict Nexus and Humanitarian Framing of Complex Displacement Contexts*. University of Sydney. SSRN. https://doi.org/10.2139/ssrn.4884888

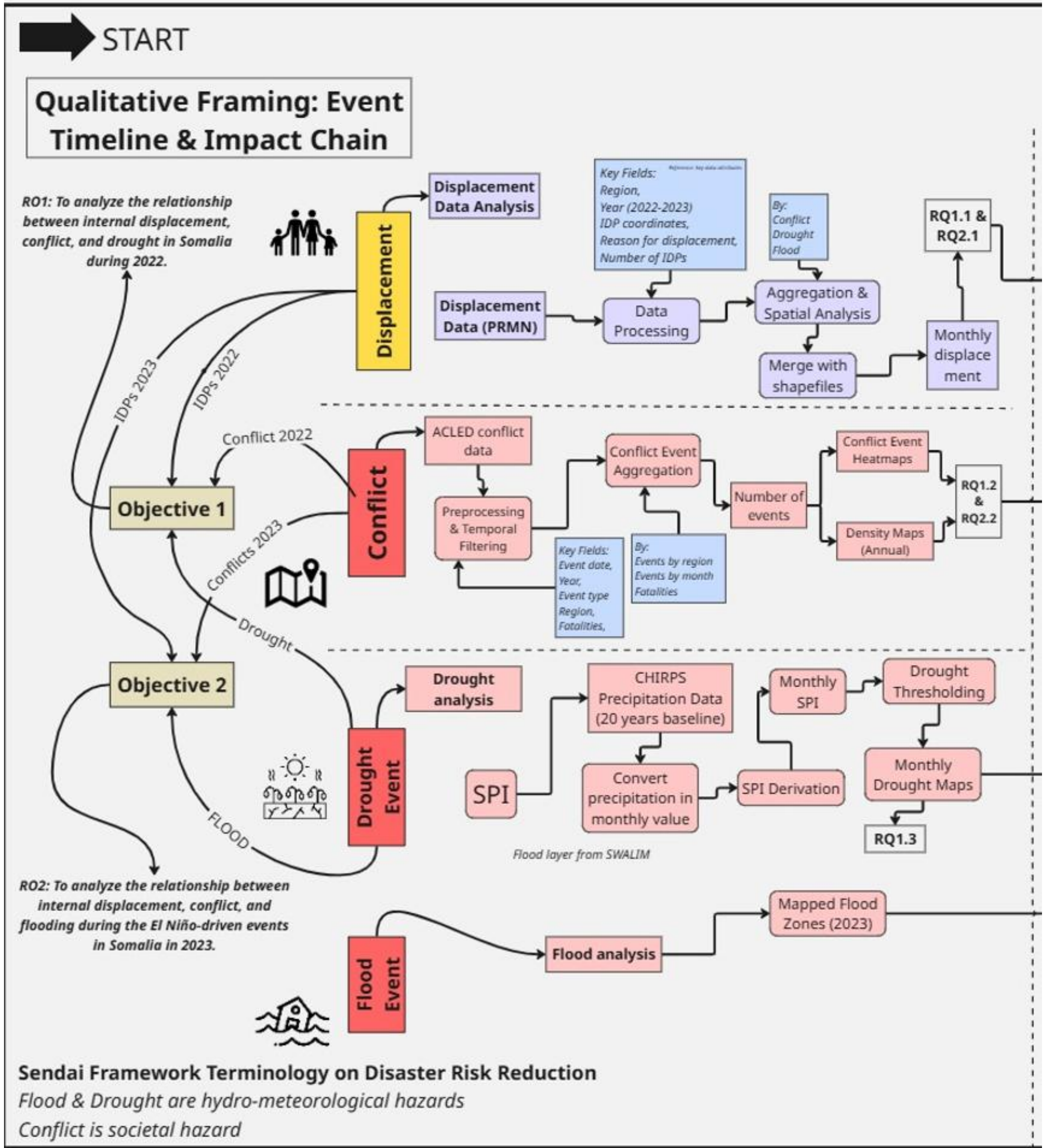
Moving forward



Anticipatory Action for Displacement

METHODOLOGY




START
Qualitative Framing: Event Timeline & Impact Chain
RO1: To analyze the relationship between internal displacement, conflict, and drought in Somalia during 2022.
Displacement
Displacement Data Analysis
Displacement Data (PRMN)
Key Fields:
Region,
Year (2022-2023)
IDP coordinates,
Reason for displacement,
Number of IDPs
Data Processing
By:
Conflict
Drought
Flood
Aggregation & Spatial Analysis
Merge with shapefiles
Monthly displace ment
RQ1.1 & RQ2.1
IDPs 2023
IDPs 2022
Conflict 2022
Conflicts 2023
Drought
FLOOD
Objective 1
Objective 2
Conflict
ACLED conflict data
Preprocessing & Temporal Filtering
Key Fields:
Event date,
Year,
Event type
Region,
Fatalities,
Conflict Event Aggregation
By:
Events by region
Events by month
Fatalities
Number of events
Conflict Event Heatmaps
Density Maps (Annual)
RQ1.2 & RQ2.2
Drought Event
Drought analysis
SPI
CHIRPS Precipitation Data (20 years baseline)
Convert precipitation in monthly value
SPI Derivation
Monthly SPI
Drought Thresholding
Monthly Drought Maps
RQ1.3
Flood layer from SWALIM
RO2: To analyze the relationship between internal displacement, conflict, and flooding during the El Niño-driven events in Somalia in 2023.
Flood Event
Flood analysis
Mapped Flood Zones (2023)
Sendai Framework Terminology on Disaster Risk Reduction
Flood & Drought are hydro-meteorological hazards
Conflict is societal hazard

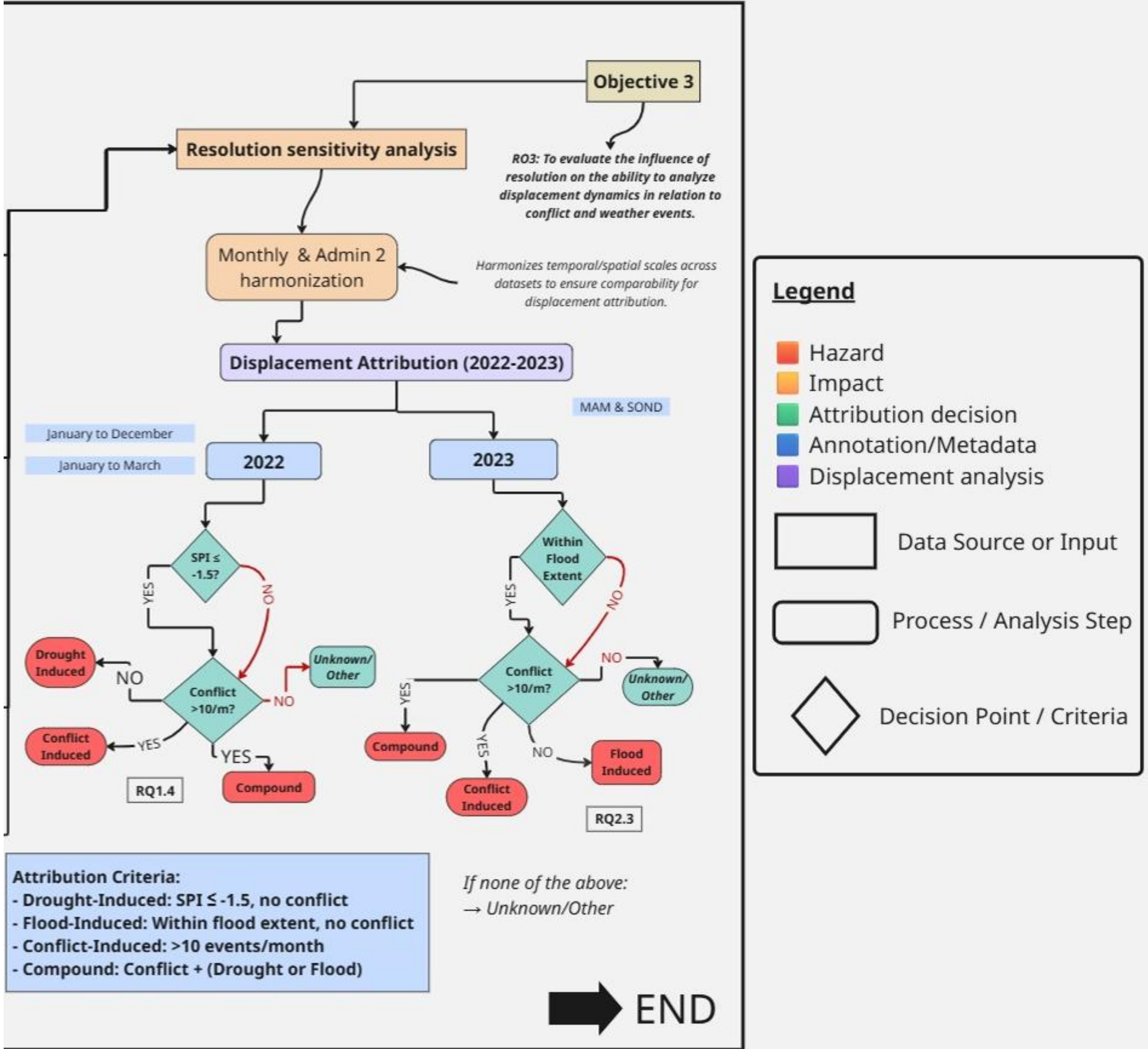

Objective 3
Resolution sensitivity analysis
RO3: To evaluate the influence of resolution on the ability to analyze displacement dynamics in relation to conflict and weather events.
Monthly & Admin 2 harmonization
Harmonizes temporal/spatial scales across datasets to ensure comparability for displacement attribution.
Displacement Attribution (2022-2023)
MAM & SOND
January to December
January to March
2022
2023
SPI ≤ -1.5?
YES
NO
Within Flood Extent
Conflict >10/m?
Drought Induced
Unknown/ Other
Conflict Induced
Compound
Flood Induced
RQ1.4
RQ2.3
Attribution Criteria:
- Drought-Induced: SPI ≤ -1.5, no conflict
- Flood-Induced: Within flood extent, no conflict
- Conflict-Induced: >10 events/month
- Compound: Conflict + (Drought or Flood)
If none of the above:
→ Unknown/Other
END
Legend
Hazard
Impact
Attribution decision
Annotation/Metadata
Displacement analysis
Data Source or Input
Process / Analysis Step
Decision Point / Criteria

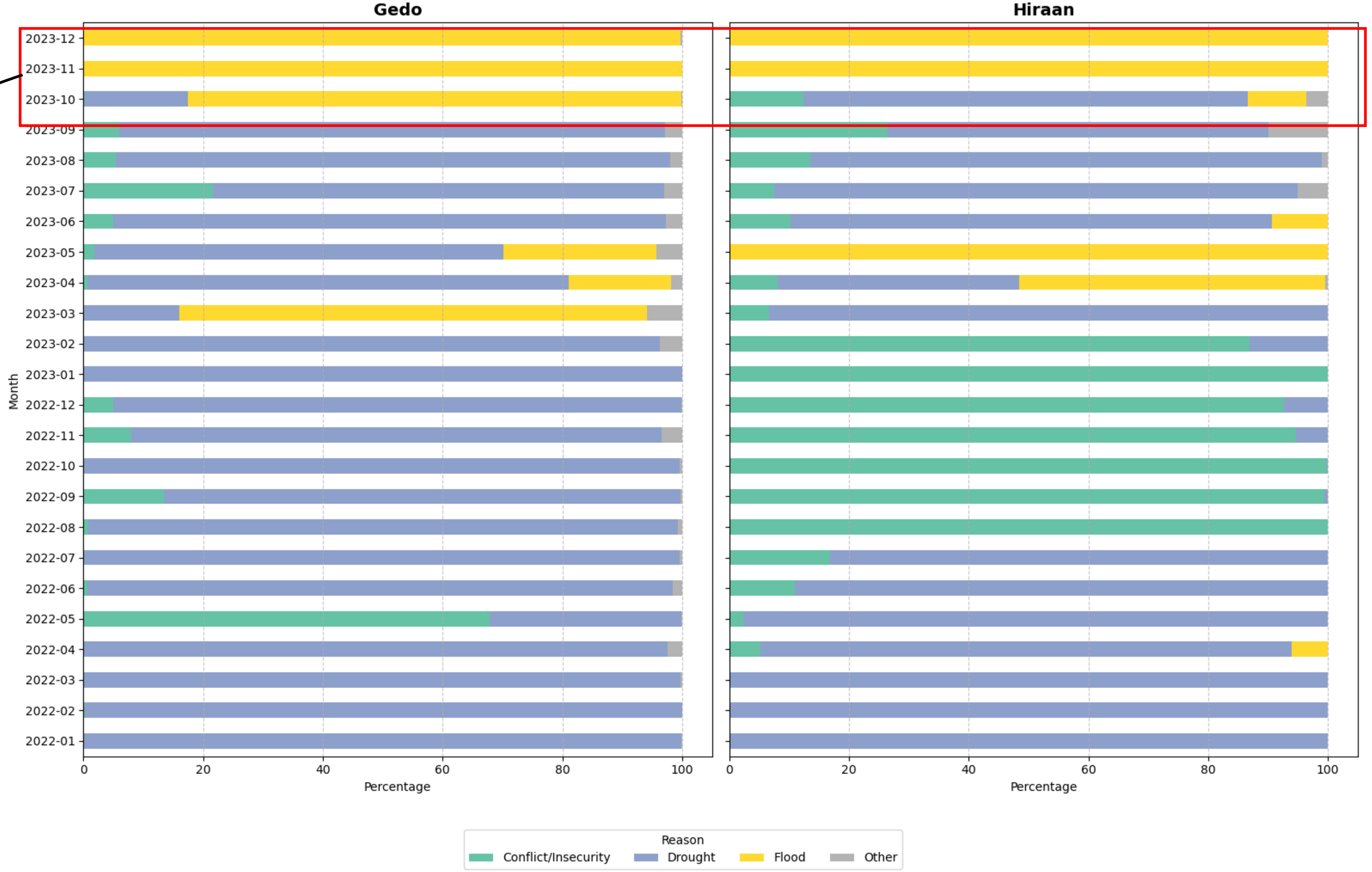
Table 4: Monthly conflict events per region in 2022

Region	01	02	03	04	05	06	07	08	09	10	11	12	Total
Awdal	2	1	1	0	0	0	0	0	0	1	2	1	8
Bakool	12	6	8	10	4	3	17	11	8	11	10	8	108
Banadir	32	33	32	36	41	22	25	39	28	46	21	42	397
Bari	4	8	7	1	4	7	4	4	3	2	2	3	49
Bay	17	23	16	23	14	25	13	34	31	18	25	24	263
Galgaduud	23	7	11	9	12	10	12	19	33	12	18	18	184
Gedo	9	4	6	7	5	15	8	8	9	8	6	4	89
Hiraan	21	16	28	18	24	21	18	33	62	26	25	28	320
Lower Juba	26	26	35	37	28	22	26	47	40	35	41	26	389
Lower Shabelle	76	82	60	102	127	102	89	104	95	92	103	113	1145
Middle Juba	1	2	0	1	0	0	1	0	2	1	2	1	11
Middle Shabelle	28	16	18	27	20	22	17	14	14	49	36	33	294
Mudug	5	4	8	7	5	2	6	1	10	8	3	7	66
Nugaal	3	1	0	3	0	3	2	1	0	3	2	2	20
Sanaag	1	2	1	0	0	1	0	2	0	1	2	0	10
Sool	2	2	1	3	4	2	0	3	0	2	2	9	30
Togdheer	2	0	1	1	1	4	1	0	0	2	3	0	15
Woqooyi Galbeed	1	3	1	3	4	1	3	0	1	2	4	2	25

SOCIETAL RELEVANCE

Source: The Sustainable Development Goals (SDG) 2015

Displacement Reasons Gedo & Hiiraan (2022-2023)

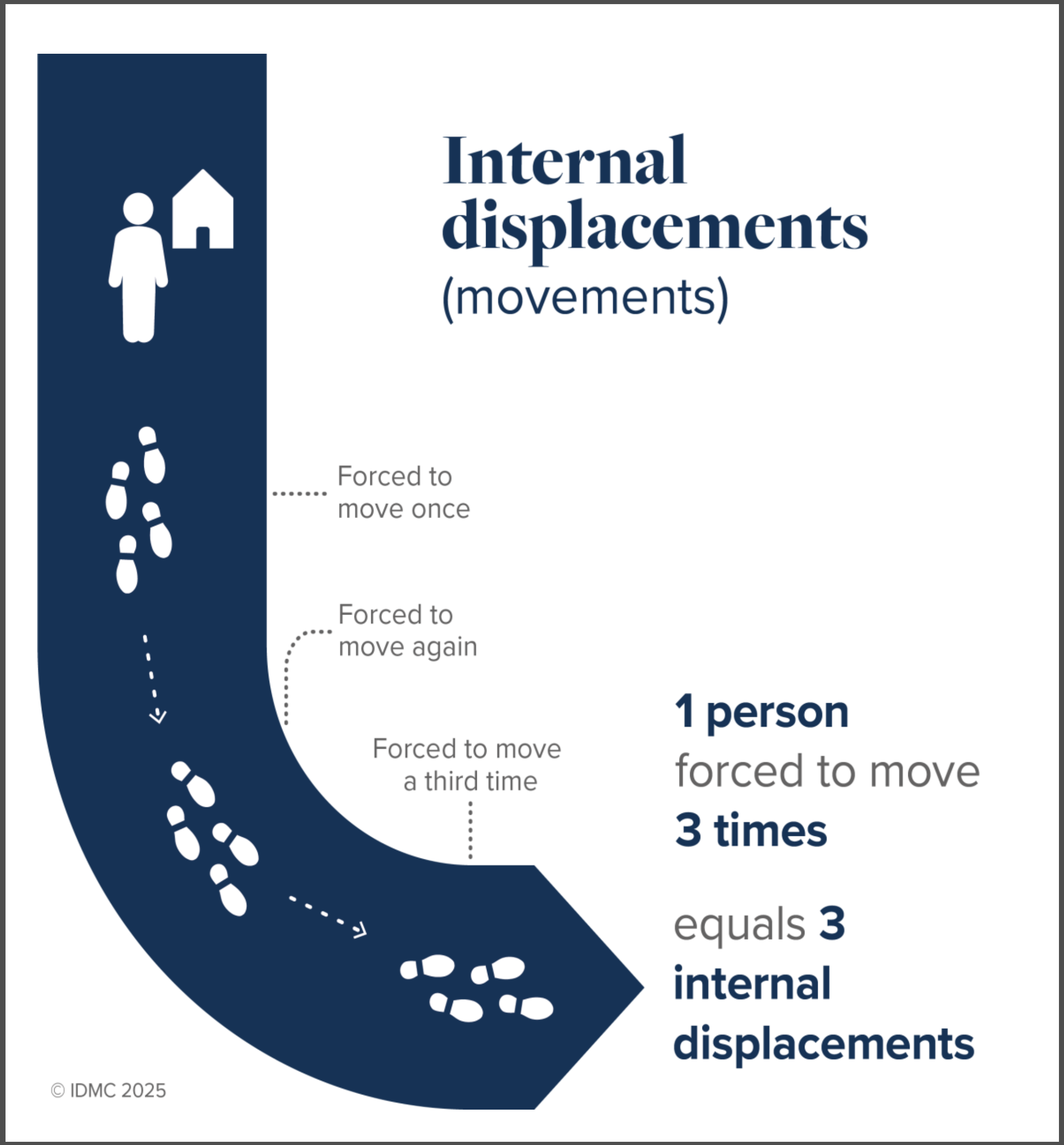


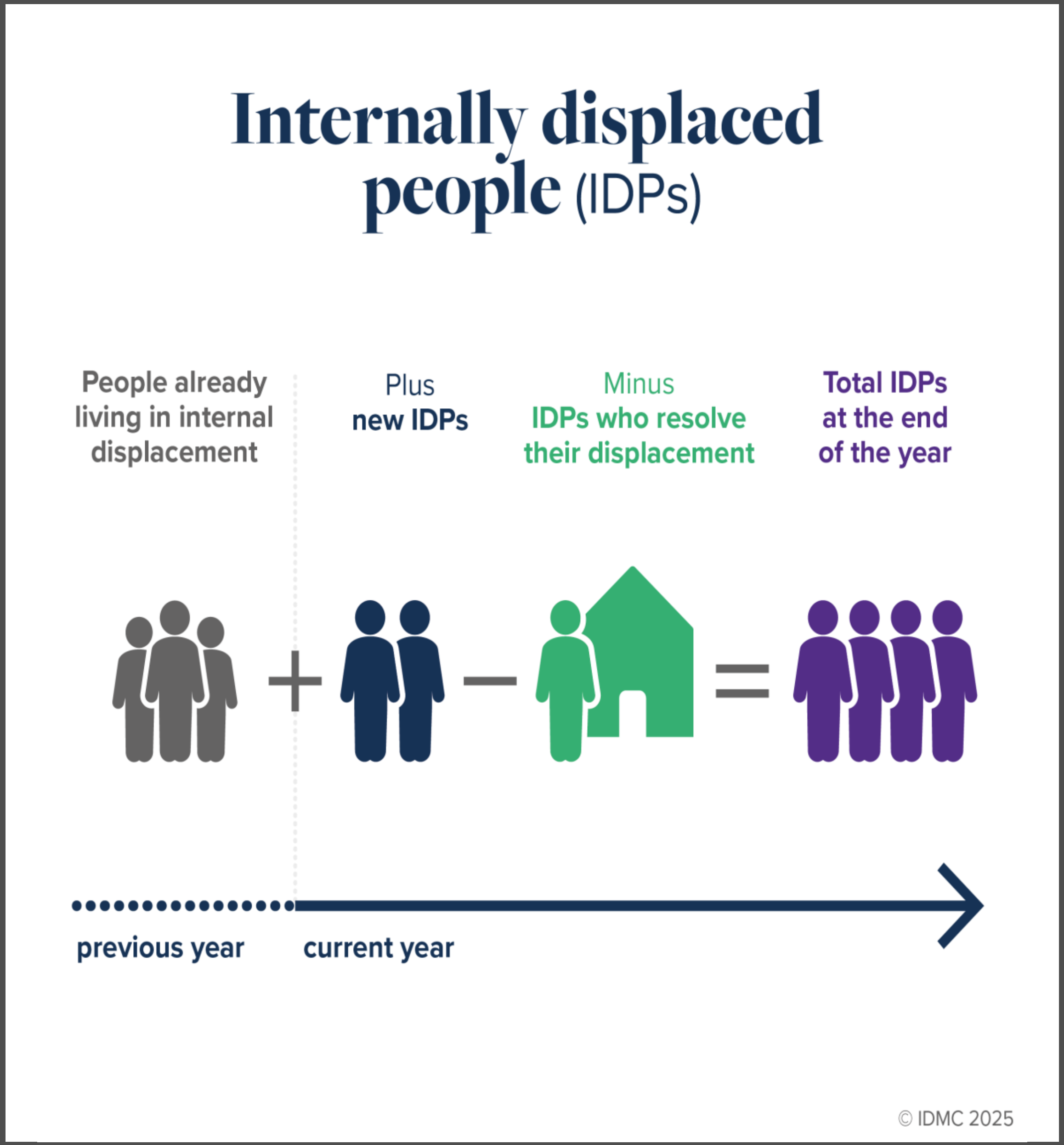
Flood-induced

Gedo

338,324 (Nov)

Validation with PRMN interviews


Internal displacements
(movements)
Forced to move once
Forced to move again
Forced to move a third time
1 person forced to move 3 times
equals 3 internal displacements
© IDMC 2025


Internally displaced people (IDPs)
People already living in internal displacement
Plus new IDPs
Minus IDPs who resolve their displacement
Total IDPs at the end of the year
previous year
current year
© IDMC 2025

Why rainfall patterns influence displacement in Somalia

1. Drought means below-average rainfall, **but not all rainfall deficits cause displacement.**
2. Failed rainy seasons reduce crop yields and pasture availability.
3. Leads to food insecurity, water scarcity, and income loss.
4. **Households may migrate seeking alternative livelihoods or aid.**
5. Coping capacity weakens with repeated shocks like consecutive failed seasons.
6. Livestock losses, debt, and exhausted social support push families to relocate.
7. Displacement is a last survival strategy, not just a direct response to the hazard.

LIMITATIONS

a. Displacement Cause Attribution

- UNHCR data records **only one primary cause** per displacement event
- Underlying drivers like **climate change, fragility, prior shocks** may be missed

b. Limited Exploration of Return & Secondary Displacement

- Analysis focused on **displacement only**
- No data on **returnees, repeated or prolonged displacement**
- Limits understanding of the **full displacement cycle**

c. Temporal Scope & Representativeness

- Focused on **2022–2023** (drought + flooding)
- Captures **short-term shocks**, but not long-term displacement patterns
- May miss cumulative effects of **historical droughts/conflicts**

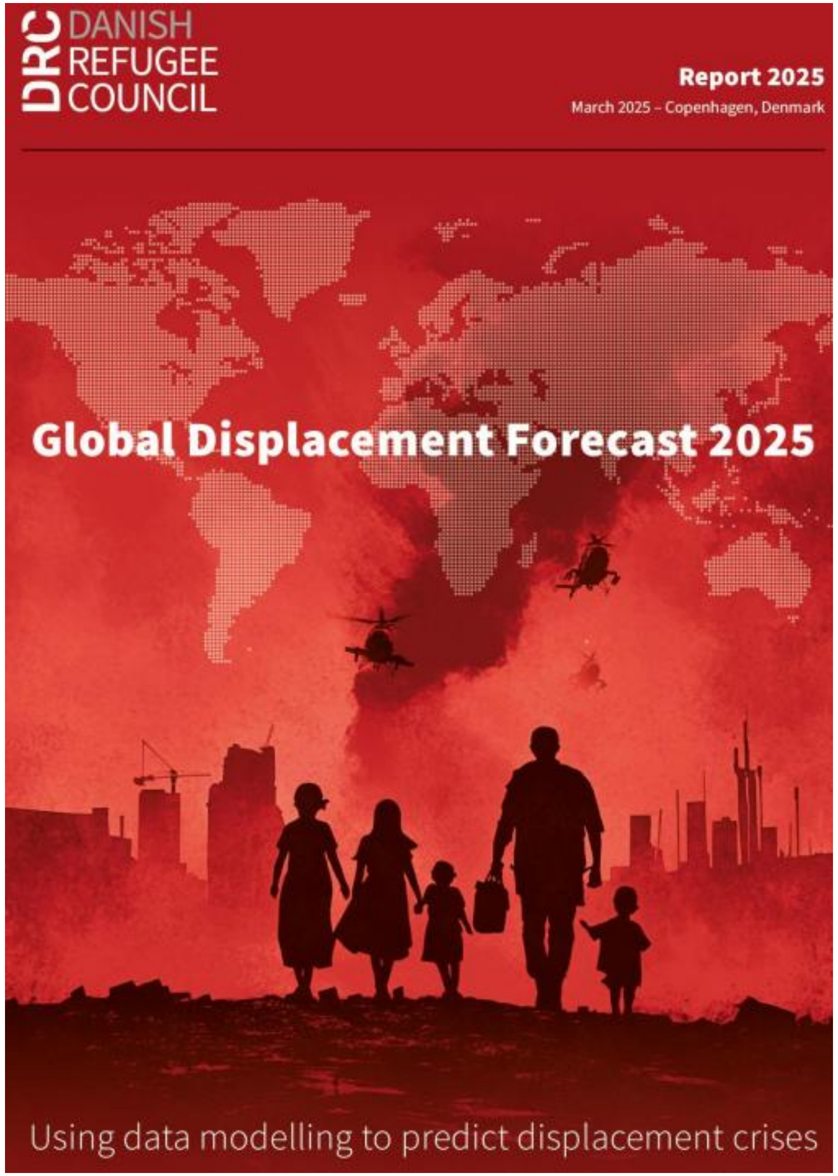
d. Conflict Threshold Sensitivity

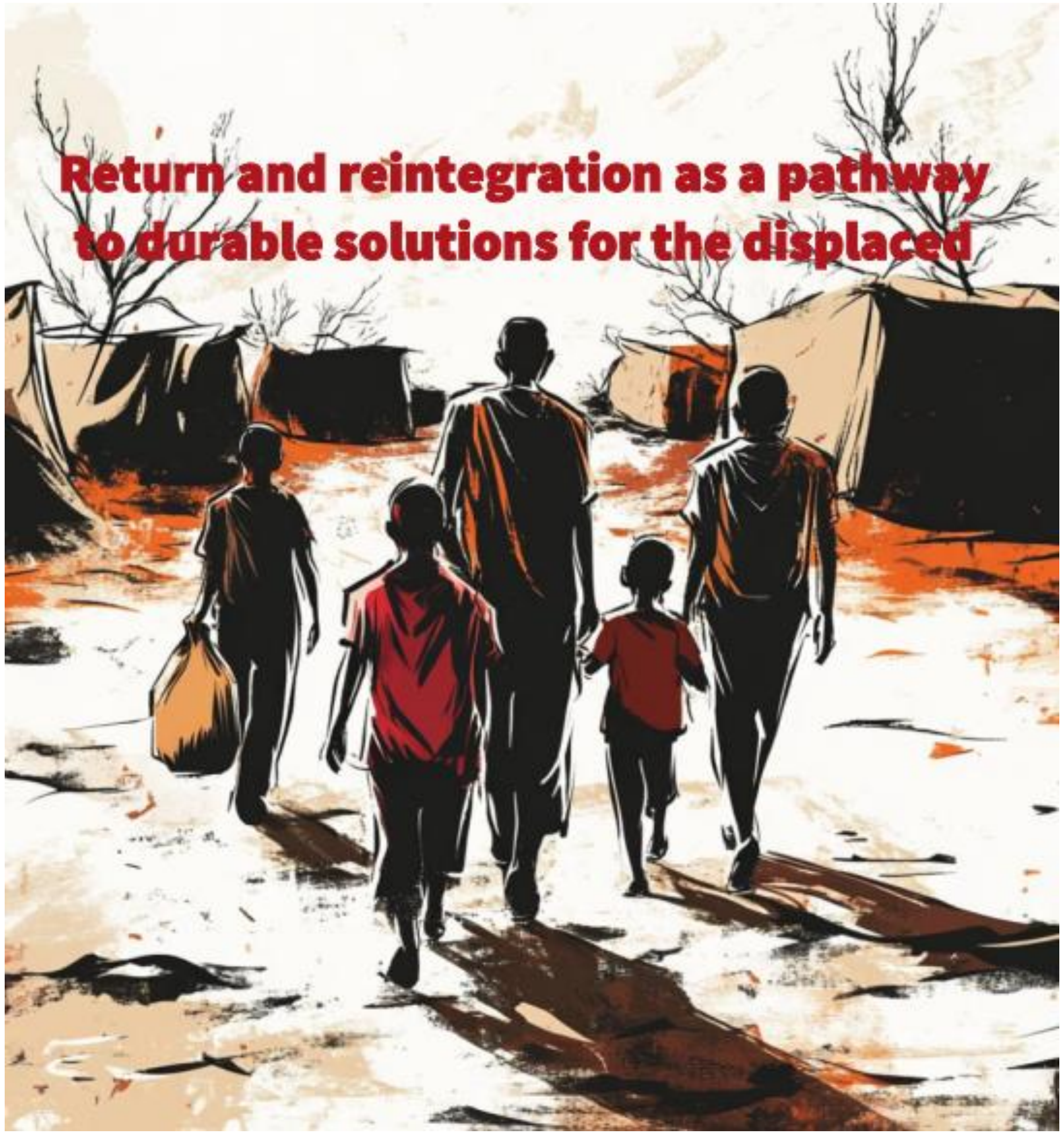
- Used **threshold-based approach** to define high-conflict regions
- Conflict intensity varies—**low-frequency events** can still cause major displacement
- Attribution results may be **context-specific** to Somalia; limited scalability

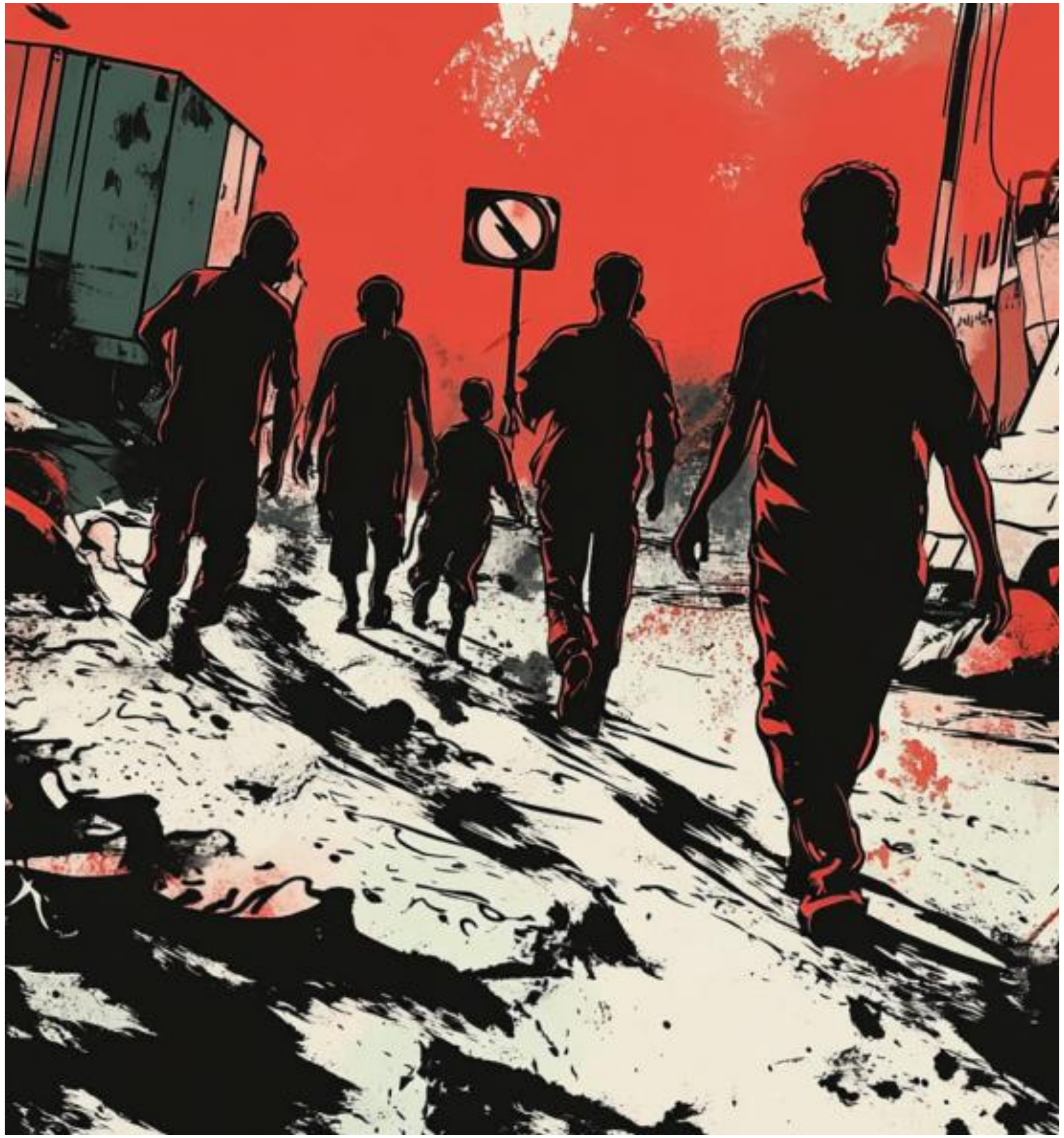
<table>
<tr><th>Displacement Analysis</th><th>Hazard Analysis</th><th>Attribution & Resolution</th></tr>
<tr><td>PRMN Displacement Data
• Monthly, Admin 2
• IDP cause + numbers
Data Cleaning & Aggregation</td><td>Conflict Events (ACLED)
• Filtering by date & fatalities

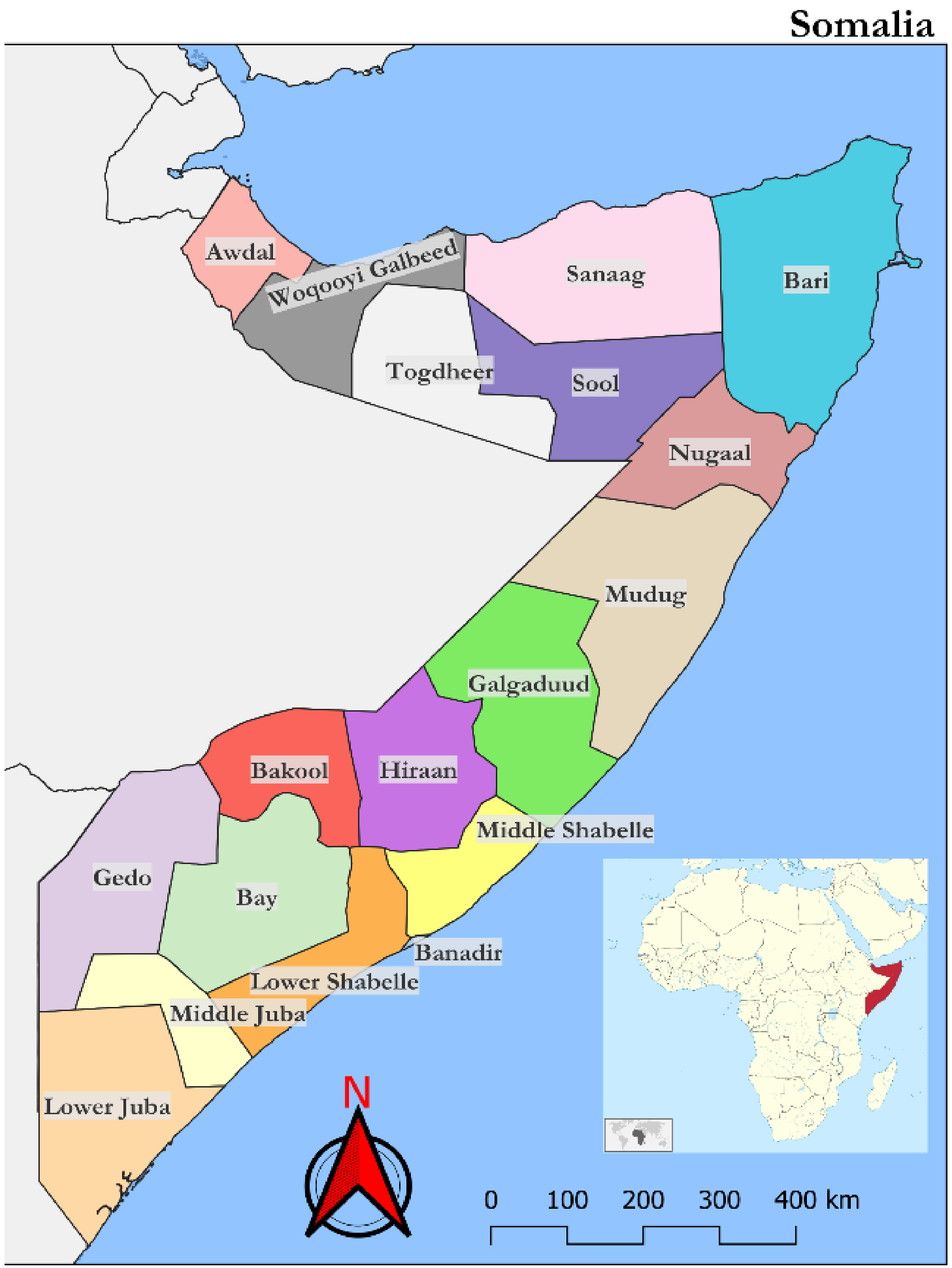
Drought & Flood Data
• CHIRPS rainfall & SPI maps
• SWALIM flood zones → Buffer</td><td>Monthly/Admin 2 Harmonization
→ All datasets aligned spatially & temporally

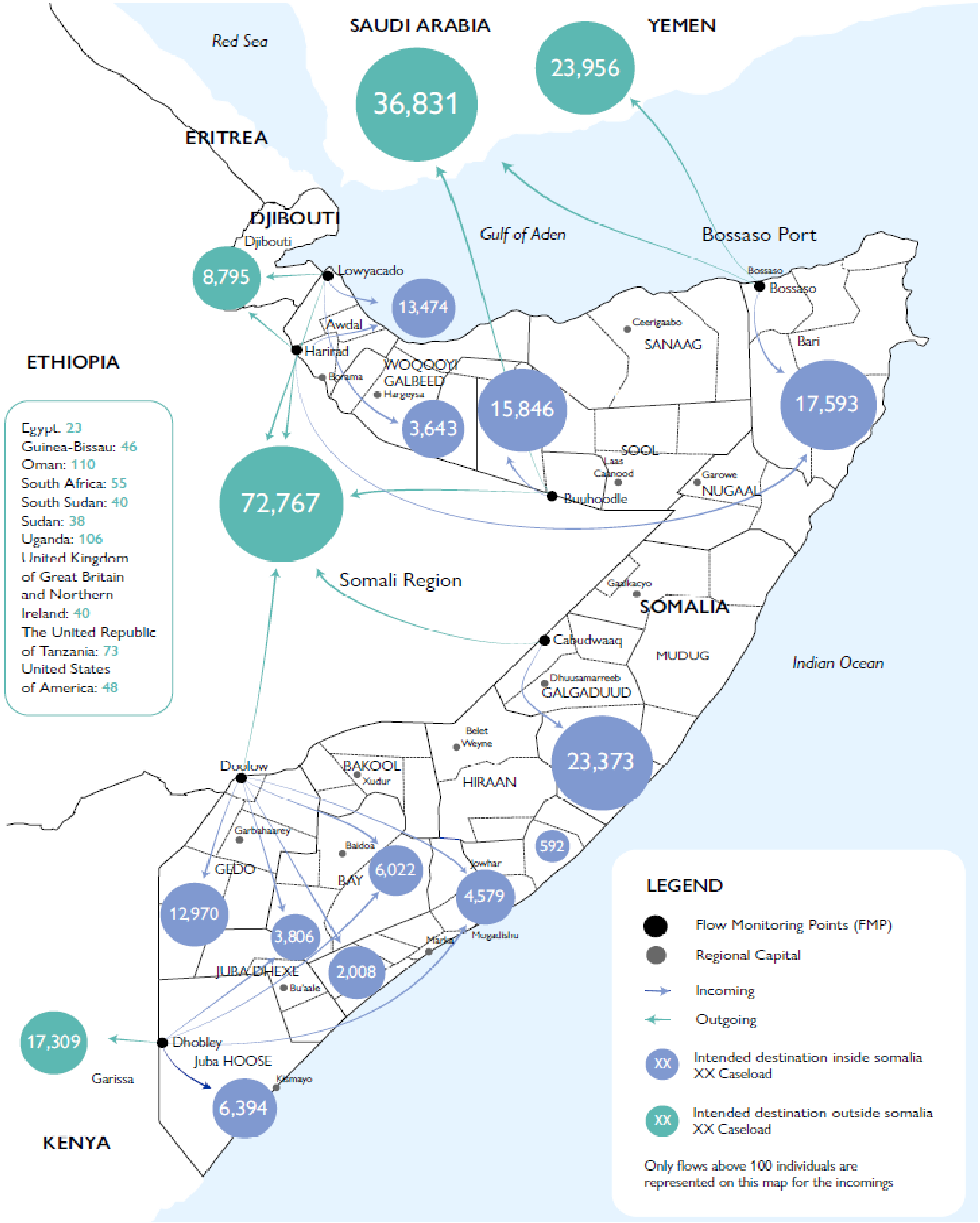
Displacement Attribution
• Drought: SPI ≤ –1.5, no persistent conflict
• Flood: within flood extent buffer
• Conflict: >10 events/month
• Compound: combination</td></tr>
<tr><td>Output:
Monthly Displacement per Region & Month</td><td>Output:
Conflict Events, Drought/Flood Maps</td><td>Output:
Displacement Attribution Tables (2022–2023)</td></tr>
</table>


DRC DANISH REFUGEE COUNCIL
Report 2025
March 2025 – Copenhagen, Denmark
Global Displacement Forecast 2025
Using data modelling to predict displacement crises


Return and reintegration as a pathway
to durable solutions for the displaced





Somalia
Awdal
Woqooyi Galbeed
Sanaag
Bari
Togdheer
Sool
Nugaal
Mudug
Galgaduud
Bakool
Hiraan
Middle Shabelle
Gedo
Bay
Banadir
Lower Shabelle
Middle Juba
Lower Juba
N
0 100 200 300 400 km


Red Sea
SAUDI ARABIA
YEMEN
36,831
23,956
ERITREA
DJIBOUTI
Djibouti
Gulf of Aden
Bossaso Port
8,795
Lowyacado
13,474
Awdal
Harirad
WOQOOYI GALBEED
Hargeysa
Borama
Ceerigaabo
SANAAG
Bossaso
Bari
ETHIOPIA
3,643
15,846
17,593
SOOL
Laas Caanood
Garowe
NUGAAL
Buuhoodle
Egypt: 23
Guinea-Bissau: 46
Oman: 110
South Africa: 55
South Sudan: 40
Sudan: 38
Uganda: 106
United Kingdom of Great Britain and Northern Ireland: 40
The United Republic of Tanzania: 73
United States of America: 48
72,767
Somali Region
Gaalkacyo
SOMALIA
Cabudwaaq
MUDUG
Indian Ocean
Dhuusamarreeb
GALGADUUD
Belet Weyne
Doolow
BAKOOL
Xudur
HIRAAN
23,373
Garbahaarey
GEDO
Baidoa
BAY
6,022
592
Jowhar
4,579
12,970
3,806
Mogadishu
Marka
JUBA DHEXE
Bu'aale
2,008
17,309
Garissa
Dhobley
Juba HOOSE
Kismayo
6,394
KENYA
LEGEND
Flow Monitoring Points (FMP)
Regional Capital
Incoming
Outgoing
Intended destination inside somalia XX Caseload
Intended destination outside somalia XX Caseload
Only flows above 100 individuals are represented on this map for the incomings